

OUTSIDE OPTIONS, COERCION AND WAGES: REMOVING THE SUGAR COATING*

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In economies with a large informal sector firms can increase profits by reducing workers' outside options in that informal sector. We formalise this idea in a simple model of an agricultural economy with plantation owners who lobby the government to enact coercive policies—e.g., the eviction and incarceration of squatting smallhold farmers—that reduce the value to working outside the formal sector. Using unique data for 14 British West Indies 'sugar islands' from 1838 (the year of slave emancipation) until 1913, we examine the impact of plantation owners' power on wages and coercion-related incarceration. To gain identification, we utilise exogenous variation in the strength of the plantation system in the different islands over time. Where planter power declined we see that incarceration rates dropped, and agricultural wages rose, accompanied by a *decline* in formal agricultural employment.

The fact that the wage level in the capitalist sector depends upon earnings in the subsistence sector is sometimes of immense political importance, since its effect is that capitalists have a direct interest in holding down the productivity of the subsistence workers. Thus the owners of plantations [. . .] if they are influential in the government, will not be found using their influence to expand the facilities for agricultural extension. They will not support proposals for land settlement, and are often instead to be found engaged in turning the peasants off their lands.

—Lewis (1954)

Many economists and historians would agree with Acemoglu and Wolitzky's (2011) assessment that 'the majority of labour transactions throughout much of history and a significant fraction of such transactions in many developing countries today are coercive'. Indeed, labour coercion is at the heart of much of the literature on long-run development and institutional change (Domar, 1970; Engerman, 1999; Acemoglu *et al.*, 2001; 2002; Engerman and Sokoloff, 2002; Greif, 2005; Nunn, 2008; Dell, 2010; Naidu, 2010; Nunn and Wantchekon, 2011; Naidu and Yuchtman, 2013; Bobonis and Morrow, 2014; Ashraf *et al.*, 2020; Lowes and Montero, 2018). Despite

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The data and codes for this paper are available on the Journal website. They were checked for their ability to replicate the results presented in the paper.

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this, rigorous empirical evidence on labour coercion is scarce and mostly focused on relating present-day outcomes to historical labour coercion.¹

Focusing on the workings of labour coercion rather than its long-run consequences, we exploit a historical setting involving 14 British West Indies sugar colonies from 1838—the year slaves were emancipated in the British Empire—to 1913. We are thus studying 14 free labour markets at their inception. Before 1838, the 14 colonies we study were exceedingly similar. Economically, all were slave societies and all were completely specialised in sugar cane production. Institutionally, all had the same political and legal systems inherited from Britain and were dominated by a small group of white planters. After emancipation, ‘the main fact of life in the free West Indies was that black labourers were unwilling to remain submissive and disciplined cane workers’ (Green, 1976, p. 170). We study how planters over the ensuing 76 years used their influence over the state to enact coercive policies that kept wages low and secured a steady supply of labour.

In keeping with Lewis’s quote, our focus is on ‘legal coercion’, i.e., the use of the state’s legislative and judicial institutions to manipulate workers’ outside options. This focus is particularly pertinent where workers’ wages in the formal sector are determined by outside options generated in the informal sector. Our paper has two empirical objectives: one, we want to test to what extent legal coercion was used to lower wages. Two, we study the importance of the planters’ economic and political influence over the government in shaping legal coercion.

To guide our empirics, we employ a simple model where workers earn a wage w that is equal to their outside option in the informal sector. Coercion C reduces this outside option, e.g., by evicting smallholders from their plots. Coercive policies C are set by the government to benefit the planters as in Grossman and Helpman’s ‘Protection for Sale’ framework (1994), and the government chooses a higher C when planters’ influence is greater.² Planter influence depends on the number of plantations on the island (N), which in turn depends on the extent to which Caribbean plantations offer higher returns to British investors than returns obtainable elsewhere. In our model a shrinking plantation sector leads to lower levels of coercive activity which raises wages: $N \rightarrow C \rightarrow w$. Exogenous factors act as an instrument for N .

The data are collected from the British Colonial Office’s *Blue Books*: wages w_{it} are reported as the annual average of the agricultural spot market wage. C_{it} is measured by incarceration rates per capita. N_{it} is measured as the share of sugar in total exports or as the share of all plantation crops in total exports. Our empirical focus is on within-colony over-time panel variation.

Our hypothesis is that where the plantation system went into decline, C_{it} decreased and w_{it} increased relative to other islands. A generalised difference-in-differences strategy robustly bears this out across a range of different specifications. We instrument for N_{it} using exogenous variation in the returns to investing in Caribbean plantations relative to investing elsewhere. *Across islands*, returns in the Caribbean were lower where smallholders could easily escape work on plantations. Some colonies had large hinterlands while others had next to none.³ *Over time*, returns to investing outside of the Caribbean improved rapidly over the nineteenth century. As the Empire

¹ Two exceptions are Naidu and Yuchtman (2013) and Bobonis and Morrow (2014), summarised below.

² One may think of planter influence as lobbying capacity but this is not explicitly modelled. Our own view is that legal coercion is in practice almost always the result of collective action by an elite that influences the state to regulate coercive labour laws to their benefit, e.g., to reduce rights at work, to harass workers in the informal sector or to limit worker mobility, with the ‘Black Codes’ in the post-bellum US South being a prominent example. In Section 1, we describe in detail the practical applications of legal coercion in our context.

³ During slavery, this difference did not matter: land that was unsuitable for sugar lay uncultivated even if it was very fertile. This basic explanation of the divergent post-emancipation fortunes of ex-slaves across the islands figures prominently in the literature on Caribbean history and indeed it was anticipated in the 1830s debates surrounding

expanded, opportunities to export British manufactures grew much faster than opportunities in Caribbean agriculture, and English investors turned their attentions away from the Caribbean. Combining these exogenous elements, our instrument O_{it} is the product of the cross-section of islands' hinterlands and the time-series of English exports to non-Caribbean destinations. In the data, a higher O_{it} led to a decline in N_{it} and C_{it} and a rise in wages. When we instrument for N_{it} we get two-stage least-squares (TSLS) estimates that are highly statistically significant and economically large: a 50% decline in the plantation system—roughly the average over the 76 years we study—would have increased wages by about 50% and reduced incarceration rates by close to their mean of 1 in 100 people. These results hold up under an extensive set of robustness checks.

We also provide evidence on the mechanisms linking N , C and w . A causal *mediation analysis* shows that three-quarters of the negative effect of the plantation system on wages operated via measured coercion, *prima facie* evidence for our hypothesis. To lend support to our assumption that coercion was 'legal coercion' exercised by the government in response to pressure from the plantation system, we study the effects of an exogenous shock which changed the composition of the plantation-owning elite, while keeping the size of the plantation sector constant. This shock strengthened the plantation system's lobbying power with each island's government apparatus, but severed the localised ties between the previous plantation owners and the parochial constabulary and judges. We find that this shock increased coercive legislation, which was set centrally, but decreased incarceration, which depended on these parochial ties. This evidence suggests that legal coercion adjusted when the planters' form of influence changed.

We conclude this introduction with a literature review. The importance of using the legislative, judicial and policing powers of the state to reduce the outside options of workers and to benefit a small elite is emphasised by Lewis (1954) and also in Acemoglu and Robinson's (2012) study of apartheid. In the focus on the manipulation of workers' outside options, we relate to Alston and Ferrie's account of Southern paternalism (1993). The closest empirical study to ours is Bobonis and Morrow (2014) who show that coffee price shocks in Puerto Rico in the mid-1800s led elites to reduce human capital investments so as to depress plantation workers' outside options. We also connect to a large literature on labour coercion. This literature is more focused on the coercion of workers on the job (Chwe, 1990; Basu, 1999; Bloch and Rao, 2002; Naidu and Yuchtman, 2013); however, in a principal-agent framework, there is a complementarity between coercing workers on the job, and reducing their outside options as non-workers (Acemoglu and Wolitzky, 2011).⁴ In our setting, the latter form of coercion was clearly dominant.

In an interesting counterpoint to our findings, Dell and Olken (2020) show that the location of Dutch colonial sugar plantations on the island of Java had positive long-run effects on local infrastructure and economic development. This line of research suggests that there are potentially positive and non-institutional long-run consequences of colonial extraction. Lastly, our focus on the political consequences of geographic variation in the ability to evade the plantation system connects us to a broader literature on the institutional consequences of geography; see, e.g., Acemoglu *et al.* (2002); Engerman and Sokoloff (2002).⁵

emancipation, see Merivale (1861, pp. 312–7), Engerman (1984, p. 137), Richardson (1997, pp. 134–5, 157–8) and Patterson (2013).

⁴ Lowering wages in the bad state in order to induce effort (i.e., relax workers' incentive compatibility constraint) is easier if legal coercion simultaneously prevents the worker from walking away (i.e., relaxes the participation constraint).

⁵ Lastly, we naturally connect to a large literature on the Caribbean's economic adjustment to emancipation (Eisner, 1961; Engerman, 1982; 1984).

1. History

British West Indies plantations were a source of vast profits for British planters and investors. These profits were dealt a severe blow by the emancipation of slaves in 1838. Emancipation initially led to sharply rising wages as freed slaves rejected plantation life in favour of squatting on abandoned estates or small plots in the hinterland. This ‘flight off the estates’ did not last long. Within a few years of emancipation, the white planter elite had developed a system of legal coercion over labour that lowered wages and slowed the demise of the plantation system. We now describe the workings of this system.⁶

1.1. *Legal Coercion C_{it}*

Legal coercion in our setting took three main forms. First, coercion restricted access to land. The full force of the law was brought to bear on peasants who attempted to squat on abandoned estates or Crown land. Squatting was so rampant that it seriously undermined the ability of planters to keep peasants on plantations. In Jamaica there were 10,000 squatters by 1844 and this number probably climbed to 40,000 by the mid-1860s (Eisner, 1961, pp. 215–6). The *Colonial Blue Books* list the titles of all colonial statutes and a quick perusal shows that every colony repeatedly enacted and strengthened trespass and vagrancy laws in order to prevent squatting. The salience of the squatting-incarceration issue is illustrated by Jamaica’s Morant Bay Rebellion, which left 600 dead and many more imprisoned (Underhill, 1895; Craton, 1988).⁷ Other examples abound: in Dominica, the ‘Queen’s Three Chains’ unrest of 1856 was driven by a dispute over whether several black families had implicit title to or were squatting on the land they were farming. It was resolved by the rapid dispatch of troops from nearby Antigua (Honchurch, 1984, pp. 136–8). The ‘Toll Bar Riot’ and the ‘Florence Hall Riot’ in Jamaica were triggered respectively by objections over limiting smallholders’ access to water sources, and over the sentencing of squatters for trespassing on an abandoned estate. See Cundall (1906, pp. 5–12).⁸ A feature of land conflicts in the post-emancipation Caribbean was that they resulted from planters’ attempts to reduce peasant smallholders’ outside options. Planters were not grabbing the land for themselves. In fact, there was plenty of fallow plantation land during the period we study. For example, in mid-century Jamaica, less than one-quarter of all land was under cultivation and, even on the active plantations, less than half of the plantation land was in use. That is, disputes were not over land that could be or ever was used for plantation crops. See Satchell (1990, ch. 4 and especially, pp. 63–4). Land conflict in our case was thus different from historical episodes in other parts of the world where planters cultivated land stolen from peasants. See for example Sánchez *et al.* (2010) on Colombia’s land conflicts 1850–1925 and Acemoglu and Robinson (2012, ch. 9) on Guatemalans’ coffee land grabs during the same period.

⁶ See Merivale (1861, pp. 340–1), Engerman (1984, p. 134 and table 2) and Riviere (1972, p. 13). On the ‘flight’, see Hall (1978, p. 7), Engerman (1982, p. 199) and Green (1976, pp. 174–5, 198).

⁷ By 1865, a number of villages had been established illegally on Crown lands in the hills above Morant Bay. Tensions ran high as the government sought to limit further expansion of these villages. Things came to a head during a trespass case involving a villager who had been pasturing on an abandoned estate (Underhill, 1895, p. 59). A crowd gathered at the courthouse, violence broke out, and then quickly ignited all of Jamaica.

⁸ Planters also lobbied for a host of restrictions which limited worker access to affordable land with clear legal title. Large tracts of Crown land were kept off the market, made available only at artificially high prices, or sold only in large lot sizes, e.g., Craton (1997, pp. 390–3). For example, 83% of Trinidad’s landmass was owned by the Crown, yet was kept off the market for decades after emancipation (Sewell, 1861, pp. 103, 106, 133). Also, in many colonies peasants were prohibited from pooling their resources to buy plantations and bankrupt planters were pressured not to sell to smallholders (Eisner, 1961, p. 211, Craton, 1997, p. 390).

Second, coercion was manifest in asymmetric terms of employment contracts. If a worker started employment without a formal contract the laws of many colonies stated that he or she had implicitly entered into a one-month contract. Failure to work during that month was a ‘breach of contract’ that resulted in fines or imprisonment (House of Commons Parliamentary Papers, 1839b, pp. 205–6). If in addition to wages a worker also received a cottage and a plot for growing crops, he or she was obligated to work on the plantation for a full year. The law allowed the planter to evict a peasant for absenteeism (‘breach of contract’) and threats of eviction were very effective in forcing peasants back to work.⁹ Laws additionally allowed planters to burn or confiscate cottages and crops of evicted peasants.¹⁰ The resulting destruction led to retribution by peasants who could then be sentenced to lengthy imprisonment for ‘malicious injury to property’.

Third, coercion was manifest in a tax system that penalised smallholders. This was most apparent in the biased taxation of imports. Planters imported flour and rice to feed plantation workers and these imports competed directly with smallhold crops, making smallholding less attractive. High tariffs on foodstuffs were therefore ‘opposed by the estate interests since they tended to deplete labour reserves by driving workers from plantations to the hinterland, where they grew ground provisions’ (Rogers, 1970, p. 96). Green (1976, p. 186) similarly states that there was much political conflict ‘over import duties on food, [which] enticed freedmen to abandon estate labour in favour of the production and sale of provisions’. Property taxation was also biased against smallholders. A smallholder with 5 acres could pay higher taxes than a planter with 500 acres. Not only did such abusive smallhold taxes reduce the returns to smallholding, they also led to punitive loss of title. For example, Satchell (1990, ch. 4 and table 4.3) documents that 18,000 acres of Jamaican smallholds were repossessed after 1869 for failure to pay taxes. Many other discriminatory taxes have been documented, including export taxes that were higher on smallhold crops than on sugar, e.g., Underhill (1895, p. xvii). Dookhan (1975, p. 156) emphasises the importance of such regressive taxes, arguing in particular that the 1853 ‘Chateau Belair’ Riots in the Virgin Islands were caused by a new tax on cattle. In his words, ‘as cattle rearing was primarily the occupation of the rural negro population, this tax fell principally on them’.

Legal coercion was a fact of life in the British West Indies. Its role was simple: reduce the returns to smallholding so as to encourage peasants to work on the plantations for low wages. Restated in more theoretical language, legal coercion did not affect plantation workers directly; rather, it affected them indirectly by reducing their outside options.

1.2. *The Government Apparatus*

The three pillars of the law—lawmakers, judges and police—were all controlled by planters. To gauge their coercive effects, it is useful to apply Acemoglu and Robinson’s (2008) distinction between officially sanctioned *de jure* power and unofficially sanctioned *de facto* power. Planters’ *de jure* power was exercised primarily through their influence on discriminatory laws passed by legislators in the islands’ legislative assemblies.¹¹ Planters’ *de facto* power was exercised primarily through the selective application of these laws in different locations: rural magistrates

⁹ See House of Commons Parliamentary Papers (1839b, pp. 131, 134–6), Bolland (1981, p. 595), Dookhan (1975, p. 130) and Brizan (1984, p. 128).

¹⁰ These laws were repeatedly criticised by the governor of the Leewards in 1838 on the grounds that they were inequitable and unconstitutional, e.g., House of Commons Parliamentary Papers (1839b, pp. 61–2).

¹¹ Carvalho and Dippel (2016) show that planters completely dominated colonial legislatures, especially in the early years of emancipation.

did much of their work on plantations, had little legal training and were frequently former plantation overseers (McLewin, 1987, pp. 85–7). Local police were also beholden to planters. The first post-abolition laws constituting the police force ('Police Acts') stated that rural police were to be appointed by planters. The Leewards governor complained that this was unconstitutional, but found it difficult to pressure planter-dominated legislatures into changing the laws (House of Commons Parliamentary Papers, 1839b, p. 49).

1.3. *Planters' Relative Political Power N_{it}*

We are interested in the impact of the planter elite's political power N_{it} on coercion C_{it} . The previous section described C_{it} . We now describe N_{it} . Planters' political power was based on their economic power relative to the peasantry. Economic power in the Caribbean hinged on exports. Marshall (1968, pp. 253–4) concludes from his survey of the British West Indies that the period from roughly 1850 to 1900 was one of 'continuing expansion of the number of peasants and, more important, a marked shift by the peasants to export crop production'. This growing peasant participation in exports was clearly accompanied by the declining acreage of plantations, and the rising number of freeholders and squatters (Riviere, 1972, pp. 15–7). Eisner (1961, p. 235) argues that the 'increasing prosperity of the peasantry is thus seen to be mainly due to their growing share in export crops'.¹² Our measure of the relative economic power of the plantation system is thus the share of plantation crops in total exports.¹³

Sugar is consistently identified with a plantation mode of production and has often been argued to be detrimental to economic and social development, e.g., Sokoloff and Engerman (2000) and Easterly (2007).¹⁴ In the British West Indies, sugar was also by far and away the dominant plantation crop, completely eclipsing all other crops such as cotton or coffee. We therefore use the share of sugar in total exports as one measure of the strength of the planter elite relative to the peasantry.

Figure 1 displays the lowess-smoothed share of sugar in total exports by colony. It is best to focus on the two dominant features. First, in 1838 every colony was highly specialised in sugar. Second, by 1913 there were substantial cross-colony differences in sugar export shares. Colonies roughly divided into three groups. Five colonies remained heavily involved in sugar for the entire period (Antigua, Barbados, Guyana, St Kitts and Nevis). Four colonies saw sugar decline to less than half of total exports (St Lucia, Trinidad, Tobago, and Jamaica). Five colonies exited sugar entirely by the end of period (Virgin Islands, Grenada, Dominica, St Vincent and Montserrat). In Figure 1 and the econometric analysis below we use lowess-smoothed export shares because we

¹² Eisner (1961, pp. 220, 221, 234) has fine-grained data on Jamaican smallholds and peasant exports. Our own calculations show that between 1850 and 1890 the share of Jamaican exports originating from freeholds and squatters rose spectacularly from 10.4% to 39.0%.

¹³ An acreage-based measure would be an appealing alternative, but the data are only sporadically available.

¹⁴ Sugar was unequivocally a plantation crop (Engerman, 1983). However, there is not complete consensus in the literature about the factors that made it so, and such a discussion is beyond the scope of this paper. However, we conjecture that three factors are important: (i) the sugar mill was a major capital asset that was beyond the financial reach of all but the richest members of Caribbean society (Lobdell, 1996, pp. 322, 326; Marshall, 1996, p. 73); (ii) sugar must be processed within hours of harvesting so that there was always a sugar mill either on the plantation or nearby (see Higman's 2001, figure 2.5 map of Jamaican mills); (iii) labour demand during the sugar harvest was physically brutal (e.g., 90-hour work weeks) and conflicted with workers' needs to harvest their own provision grounds (Higman, 1984, pp. 182–3). These factors favoured a system of production that vertically integrated harvesting with milling at a single location (the plantation) and which, in the racialised post-emancipation period, used coercion rather than overtime pay as an incentive device, i.e., these three factors favoured a plantation system.

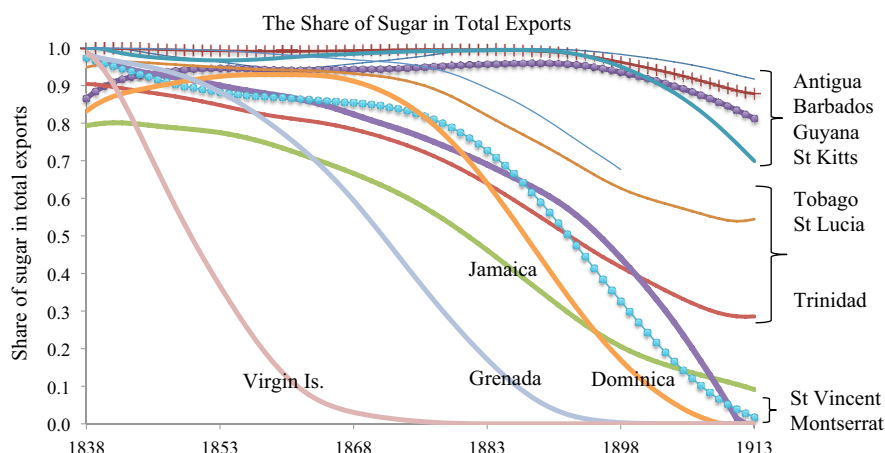


Fig. 1. *The Differential Decline of the Plantation Economy.*

Notes: This figure reports the share of sugar in total exports. Nevis is not reported because it stayed above 0.99 in each panel. Also, Nevis merged with larger St Kitts in 1883 and Tobago merged with larger Trinidad in 1899.

are interested in capturing long-run changes in the strength of the plantation system rather than short-run agricultural fluctuations.¹⁵

The data displayed in Figure 1 were intimately linked to the planter's power and influence over government, in particular their success in securing plantation labour at low wages.¹⁶ Dookhan (1977, p. 11) argues that across the Caribbean, labourer 'drift from the estates' was lowest in Antigua, St Kitts and Barbados, i.e., in precisely the three islands where the plantation system survived the longest. Green (1976, p. 258) similarly argues that the success of Barbados's plantations was 'not due to new technology but rather due to a large and disciplined labour pool'.

Over time, some planters shifted to other plantation crops, which were similar to sugar in that they also incentivised labour coercion. To handle this shift in coercive crops, we construct the share of all *plantation-produced* goods in total exports as a second measure of elite power. Using a 76-year panel of exports by colony and crop, we used historical accounts to code up the share of each crop's production accounted for by wage-paying plantations, $Pl_{git} \in [0, 1]$, and used this to calculate the share of plantation-crop exports in total exports. The detailed coding is described in Online Appendix A.¹⁷

1.4. *Exogenous Drivers of the Strength of the Plantation System*

The preceding discussion of Caribbean history suggests regressions of wages and coercion on the power of the planters. There is naturally a concern that the varying power of the planters displayed

¹⁵ The lowess smoothing faithfully reproduces the annual data. See Online Appendix Figure A.1.

¹⁶ While we do not focus on this, it is worth noting that the evolution of plantations displayed in Figure 1 also correlated tightly with the share of British whites in the Caribbean at the time. To European observers at the time, the exodus of whites was synonymous with the decline of the plantation system and the decline of the institutions that had until then characterised the British West Indies. See also Carvalho and Dippel (2016).

¹⁷ All available evidence strongly suggests that the share of other plantation crops is indeed strongly and negatively correlated with the number of smallholders and smallholder participation in exports. See, for example, Dookhan (1977, p. 136).

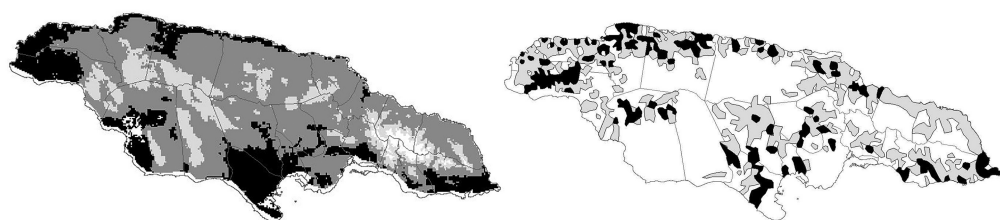


Fig. 2. *Sugar-Suitable Land in Jamaica (left), and Plantations in 1790 and 1890.*

Notes: The left panel shows the spatial distribution of land that is sugar-suitable (black), only moderately sugar-suitable (dark grey), not sugar-suitable (light grey) and totally sugar-unsuitable (white). See Online Appendix B for details. The right panel shows the extent of sugar plantations in 1790 (black plus grey areas) and 1890 (black areas). There is a good match between our estimate of sugar-suitable land in the left panel, and the historical sugar plantation land in Jamaica. The left panel is based on authors' calculations. The right panel is a digitised version of Higman's (2001) remarkable Figure 2.9.

in Figure 1 was endogenous to factors that may also have affected wages and coercion directly. It is therefore desirable to identify drivers of N_{it} that were exogenous in the sense of impacting wages and coercion only through N_{it} . Our first candidate for such a driver was a source of cross-sectional variation across colonies that became very salient to peasants after emancipation. In a colony like Barbados, all of the land was sugar-suitable and in 1838 only 4% of land was not under cultivation (1838 Barbados *Blue Book*). By contrast, a colony like Jamaica had sugar-suitable coastal plains but a higher-elevation interior (which was fertile but not sugar-suitable). During slavery, this difference between Barbados and Jamaica did not matter because sugar-suitable land was similarly suitable everywhere and the hinterland was largely inaccessible to slaves and not used by many others. After emancipation, differences in the availability of a fertile but sugar-unsuitable hinterland led to stark differences in the ease with which peasants could evade the plantation system and resist coercion (Engerman, 1984; Richardson, 1997; Patterson, 2013, respectively pp. 137, 134–5, 157–8).¹⁸

To measure this differential ability to evade the planters we carefully calculated the share of each colony's land that is unsuitable for sugar cane, O_i .¹⁹ The relationship between O_i and the historical outside options that peasants actually had is visually displayed in Figure 2 for Jamaica, the only colony with historical maps on evolving land-use patterns. In the left panel, the black areas are lands that we have coded as highly suitable for sugar cane. In the right panel, the shaded areas (both black and grey) were plantations in 1790. The two are spatially correlated and, averaging across Jamaica, the share of land under plantation in 1790 was very close to the share of highly sugar-suitable land.²⁰ This map and all the post-emancipation history make clear that O_i is a good proxy for the amount of hinterlands available to freed slaves after 1838. We note that O_i is an extremely good predictor of the long-run survival of the plantation system. This point is made visually in Online Appendix Figure B.6: the higher is O_i , the smaller is the 1913 sugar export share.

¹⁸ Engerman (1984, p. 137) quotes a Jamaican planter in the 1830s as arguing that emancipation 'will be less mischievous to other colonies than ours. For in Barbados and Antigua and several other Islands the liberated slaves must work for wages or want the necessities of life.'

¹⁹ We defer data details to Section 3 and the Appendix sections cited therein.

²⁰ The right panel of Figure 2 illustrates another point. The grey-shaded areas are plantations that shut down between 1790 and 1890. These were the lands that were most difficult to keep out of peasant hands and were thus a major focus of coercive interventions.

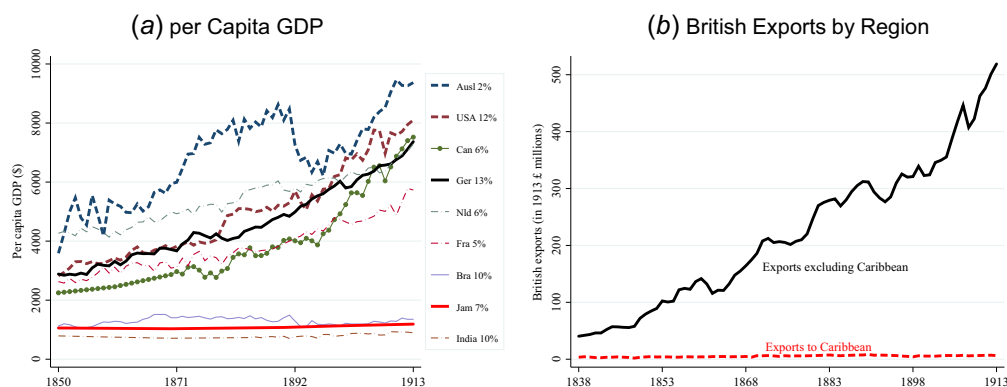


Fig. 3. *British Markets and British Exports by Destination.*

Notes: Panel (a) is per capita GDP (1990 international dollars) from Madisson, 1850–1913. (Earlier data are not consistently available.) The countries included were the top destinations for British exports in 1846 (the earliest date available for exports by country) and accounted for 71% of all British exports. The legend reports 1846 export shares, e.g., Germany received 13% of all British exports in 1846. Panel (b) is British exports to the Caribbean and British exports excluding the Caribbean. These exports are in constant 1913 pounds sterling (£). All exports exclude re-exports.

In the eighteenth century, British investors made their fortunes in Caribbean sugar. During the nineteenth century, however, investment opportunities shifted to other regions. Panel (a) of Figure 3 displays the per capita GDPs of Britain's largest export destinations at the start of our sample. Data are from an update of the Madisson data (Bolt *et al.*, 2018). The only British West Indies colony with the Madisson data is Jamaica. See the solid line at the bottom of the panel. The figure shows that while Jamaica languished, the majority of Britain's largest export destinations rapidly grew richer.

Exogenous events elsewhere around the globe were overtaking Jamaica and drawing away the attention of British investors: the repeal of the Corn Laws in 1846 gave a tremendous impetus to free trade around the world, and started what eventually came to be called the 'first globalisation', which, buttressed by a precipitous decline in ocean freight rates, lasted until 1914 (North, 1958; O'Rourke and Williamson, 2001; 2002).²¹ In this period, the Caribbean was left behind partly because the repeal of the Corn Laws included provisions for a phasing-out of preferential tariffs on West Indies sugar over the period 1846–54 (Curtin, 1954).

This globalisation-driven boon and shifting-away of British attention from the Caribbean towards the rest of the world can be seen in export volumes. Panel (b) plots British exports to the Caribbean and to all non-Caribbean destinations. Data are in constant 1913 pounds sterling (£) and purged of re-exports (e.g., Jamaican sugar that is shipped first to London and then to France). The underlying data are from Mitchell (1988) and detailed in Appendix B.5. British exports to non-Caribbean destinations display periods of rapid growth followed by shorter periods of stagnation (e.g., the 1890s) and infrequent bursts of decline. However, the overall picture is one of rapid growth, with real exports growing at an annually compounded rate of 3.4%. In comparison, British exports to the Caribbean experienced practically no growth. Judged by

²¹ Falling ocean freight rates led to a shift from trade in only high unit-value products like sugar towards trade in many lower unit value products like heavy machinery exports to the USA and Europe.

exports, then, British investors had largely turned their focus away from the Caribbean in response to exogenously improving opportunities elsewhere.

We have focused on two exogenous drivers of the strength of the plantation system that will be useful for our identification strategy. *Across colonies*, coercion on plantations was more difficult where the hinterland was large. *Over time*, the plantation system declined in part because British investors shifted their attention away from Caribbean sugar and towards opportunities in regions that were experiencing rapid growth.

1.5. Other Major Drivers of the Plantation System's Strength

1.5.1. Crop price movements

One challenge for identification is that pressures on the coercive plantation system came in part from variation in smallhold crop prices. In fact, there was a secular decline in plantation crop prices relative to smallholder crop prices in the Caribbean during this time.²² Higher smallhold crop prices would have raised smallhold exports in revenue terms and therefore reduced N_{it} , while they would have also increased smallhold profits, and by extension wages, even if planters' actual power over governance had been unaffected. We therefore carefully control for changes in the export prices of smallhold crops.²³ We also note the importance of inspecting the responses of C_{it} to N_{it} in addition to the wage responses.

We are also concerned with the fact that crop choices were endogenous, irrespective of the fact that world crop prices were plausibly exogenous to Caribbean production.²⁴ Smallholders' profits rose more if they substituted towards crops whose prices increased. The impact of the crop price changes on smallholder profits in turn was biggest in places where the crops with the best geographic suitability and the steepest price increases coincided. We therefore embed a model of crop choice into our theory, and include into our empirics an exogenous crop-suitability-driven export-price basket, estimated as in Costinot *et al.* (2016).²⁵

1.5.2. Labour supply shocks

Crop prices were not the only factor shaping plantation labour supply during this period. Fortunately, Caribbean history is quite clear on what the other big labour supply shocks were during this period: the immigration of indentured East Indian immigrants (Laurence, 1971; Riviere, 1972) and the construction of the Panama Canal (Maurer and Yu, 2013, ch. 4).

1.5.3. Hurricanes

Finally, we note that the Virgin Islands are an outlier in Online Appendix Figure B.6: the plantation system had collapsed by 1913 despite low values of O_i . Nowhere else in the Caribbean did planter power diminish as rapidly as in the Virgin Islands. As it turns out, the Virgin Islands plantation system collapsed because of two major hurricanes in 1848 and 1852, which destroyed

²² This decline in plantation prices, particularly in sugar, was important. A previous version of this paper (Dippel *et al.*, 2015) was about the impact of international trade (declining sugar prices) on wages, coercion and institutions. While we continue to carefully control for prices, they are no longer the focus of this paper.

²³ We additionally control for other important changes in plantation labour supply in the Caribbean during this time, namely the immigration of East Indian labourers and work opportunities from the construction of the Panama Canal. See Appendix B.6.

²⁴ World crop prices were exogenous to Caribbean production. Even for sugar, at their production peak in the early years of the data, the British West Indies produced only 18% of world sugar output and this number fell to 1% by the end of the sample. See Online Appendix Figure A.4.

²⁵ See Appendix B.2.

the colony's sugar infrastructure and left planters too indebted to rebuild. Hurricanes do two types of damage: they destroy crops and they destroy structures such as sugar mills. Since sugar cane must be processed within hours of harvesting and since cane is difficult to transport, there was always a sugar mill either on the plantation or nearby, e.g., Higman (2001, figure 2.5). Sugar mills were unique in Caribbean agriculture in that they were expensive and long-lived assets. They were also prone to hurricane damage. In the post-emancipation Caribbean, an increasing share of planters could only cover their variable costs but not their fixed costs. In other words, it made sense for a marginal planter to operate an existing mill, but not to rebuild a destroyed mill (Lobdell 1996, pp. 322, 326; Marshall 1996, p. 73). As a result, hurricane landfalls that destroyed mills had long-lasting effects. To control for hurricane damage, we geo-referenced the paths of every major hurricane that hit the Caribbean between 1838 and 1913, and assigned each Caribbean hurricane landfall an island-specific damage index HDI_{it} . Data sources, the list of all hurricanes and their measured impact appear in Online Appendix D.

2. A Simple Model of Coercive Labour Market Institutions

To fix ideas and provide additional motivation for the empirics, we now turn to a simple theory of coercive labour markets.

2.1. Technology and Crop Choice

There is an exogenous measure L of workers (former slaves) and an endogenous measure N of planters (members of the planter elite). There is a continuum of heterogeneous plots indexed by ω , each of which can be planted in one of $g = 1, \dots, G$ crops. We follow Costinot *et al.* (2016) in modelling crop choice by assuming that plot ω planted in crop g has a baseline yield of $z_g(\omega)$ where $z_g(\omega)$ is a random variable with a Fréchet distribution: $Pr\{z_g(\omega) < z\} = e^{-T_g z^{-\theta}}$. On a plantation, plot ω combined with one worker produces output $\tau_g^p z_g(\omega)$ where $\tau_g^p \geq 0$ describes the efficiency of plantation agriculture, e.g., τ_g^p is large for sugar and small for livestock. On a smallhold, plot ω combined with one worker produces output $\tau_g^s z_g(\omega)$ where $\tau_g^s \geq 0$ describes the efficiency of smallhold agriculture. The crop-specific τ_g^p and τ_g^s explain why some crops are better suited than others for plantation agriculture.

We consider a small open economy so that crop prices $\mathbf{p} = (p_1, \dots, p_G)$ are exogenous. Crops are chosen to solve $\max_g p_g \tau_g^j z_g(\omega)$ where $j = p$ if it is a plantation plot and $j = s$ if it is a smallhold plot. The optimal choice varies across plots, but on average the expected revenue per plot will be

$$r(\mathbf{p}, \tau^j) = \mathbf{E} \max_g p_g \tau_g^j z_g(\omega) = \left(\sum_k T_k (\tau_k^j p_k)^\theta \right)^{\frac{1}{\theta}} \Gamma, \quad j = p, s \quad (1)$$

where $\tau^j = (\tau_1^j, \dots, \tau_G^j)$ and $\Gamma = \Gamma(1/\theta - 1)$ is the gamma function. See Appendix A for the proof or Costinot *et al.* (2016, p. 215). $r(\mathbf{p}, \tau^j)$ captures how crop choices respond to prices.

Each smallholder is randomly allocated one plot and each planter is randomly allocated $l(N) \geq 1$ plots. Since each plot uses one worker, the maximum number of planters is $N = L$ and when $N = L$ each planter receives one plot, i.e., $l(L) = 1$. We also assume that the more planters there are, the more land they receive collectively ($\frac{\partial \ln l(N)N}{\partial \ln N} > 0$), but not individually ($\frac{\partial \ln l(N)}{\partial \ln N} < 0$). The latter creates a 'congestion cost' which ensures that not all agriculture is plantation agriculture.

2.2. *The Worker's Occupational Choice and Coercion*

Each smallholder must choose between plantation work and smallholding. Utility from working on the plantation is w .²⁶ Utility from smallholding is $r(\mathbf{p}, \tau^s) - C$ where C is the negative impact of planters' legal coercion on the returns to smallholding. C is endogenous. It follows that in any equilibrium with both plantation and smallhold agriculture,

$$w = r(\mathbf{p}, \tau^s) - C. \quad (2)$$

$r(\mathbf{p}, \tau^s)$ captures how wages respond to prices when crop choices are endogenous.

The costs of coercion (e.g., building jails) are given by C^γ where $\gamma > 1$. These costs are funded by a head tax on planters of C^γ/N . Consider planter profits. When there are N planters, each receives $l(N)$ plots, earns per plot revenues of $r(\mathbf{p}, \tau^p)$, pays per plot wages of $r(\mathbf{p}, \tau^s) - C$ and is left with profits of

$$\pi(C, N) = l(N) [r(\mathbf{p}, \tau^p) - r(\mathbf{p}, \tau^s) + C] - C^\gamma/N. \quad (3)$$

We use Grossman and Helpman's (1994) 'Protection for Sale' framework to determine the level of coercion C . $C \geq 0$ is chosen to maximise a weighted sum of the profits of the N planters and the L workers:

$$W(C) = \alpha N \pi + L w. \quad (4)$$

α is the weight given to planters' profits. Our key assumption is that $\alpha > 1$ so that planters have greater sway over the choice of coercion. Substituting equations (2)–(3) into (4) and maximising with respect to C subject to $C \geq 0$ yields the following characterisation of optimal coercion. Let $\bar{N}$ be the value of N for which $\alpha l(N)N - L = 0$. Under our assumptions, $\bar{N}$ is unique and $0 < \bar{N} < L$. The optimal level of coercion is

$$C^*(N) = \left(\frac{\alpha l(N)N - L}{\alpha \gamma} \right)^{\frac{1}{\gamma-1}} \quad \text{for } N \geq \bar{N} \quad (5)$$

and $C^*(N) = 0$ for $N < \bar{N}$. Since the land controlled by planters is increasing in the number of planters [$l(N)N$ is increasing in N], equation (5) implies one of our key results, namely, $C_N^* > 0$ when N is sufficiently large. The insight is simple: the stronger is the planter elite, the greater is its political influence (as measured by αN) and hence the higher is the level of coercion. Equation (5) further implies a threshold effect: when the number of planters drops below $\bar{N}$, there is no coercion. See Appendix A for proofs.^{27,28}

²⁶ By equating utility with income we are implicitly assuming that only the numeraire good is consumed and that all other goods are exported.

²⁷ We note in passing that if $\alpha = 1$ then $\bar{N} = L$ so that $C^* = 0$ for all N , which reflects the fact that coercion is an inefficient redistributive policy that would never be used if smallholders had equal say in choosing coercion.

²⁸ This 'Protection for Sale' setup abstracts away from part of the collective action problem in that the level of coercion grows with the number of planters. However, planters do not solve the bigger collective action problem, namely, that of collectively restricting entry into planting and thereby preventing profits from being driven to zero. Historically, in the median colony whites represented only 1.6% of the population so that, in the highly racial colonial society, whites 'stuck together'. Thus, empirically, there was no white collective action problem when it came to policies restricting black smallholders.

2.3. Worker Resistance and Planters' Entry Decision

As discussed in the history section, workers often resisted white rule, which resulted in deaths, property damage and incarceration. Since worker motivations for resistance do not enter into the empirics, we model resistance simply as a probability χ that resistance is successful. $\chi(O)$ is increasing in the share of non-sugar-suitable land O because it is costly for the police and military to operate in these more remote and often highland areas.²⁹ Without loss of generality we assume $\chi(O) = O$. If resistance is successful, neither planters nor workers are able to plant their crops and returns to each are normalised to 0. If resistance is unsuccessful, then workers and planters generate the earnings, profits and coercion levels that appear in equations (2)–(5).

We next turn to the free entry condition for planters. Each potential planter must choose between (1) staying in England where he can invest, export to non-Caribbean countries and earn $\bar{W}$ versus (2) moving to the colony where he can invest in sugar, export back to England and earn planter profits. Conditional on no resistance, planter profits are given by equation (3), namely, $\pi(N) \equiv \pi(C^*(N), N)$. Thus, in the colony the planter earns expected profits $(1 - O)\pi(N)$. If $(1 - O)\pi(N) < \bar{W}$ for all N then no planter moves to the colony and there is only smallholding. If $(1 - O)\pi(N) > \bar{W}$ for all N then L planters move, each has one plot and one worker, and there are no smallholders. We focus on the intermediate case where planters and smallholders coexist. In that case there is an N^* such that $(1 - O)\pi(N^*) = \bar{W}$ or

$$\pi(N^*) = \frac{\bar{W}}{1 - O}. \quad (6)$$

This equation pins down the equilibrium number of planters N^* . In Appendix A we provide sufficient conditions on the underlying parameters of the model for such an N^* to exist and to be stable in the usual sense that $\pi_N(N^*) < 0$.

In any stable equilibrium, N^* is decreasing in O and $\bar{W}$. Further, N^* depends on the interaction of O with $\bar{W}$. We exploit this interaction in the empirical work.³⁰

2.4. General Equilibrium and Comparative Statics

An equilibrium in our small open economy is a crop choice for each plot ω and mode $j = p, s$ that solves $\max_g p_g \tau_g^j z_g(\omega)$, a wage w that leaves each smallholder indifferent between plantation work and smallholding (equation (2)), a level of coercion $C^*(N^*)$ that maximises planter-biased societal welfare (equation (5)), and a mass of planters N^* that leaves each planter indifferent between staying in England and moving to the colony (equation (6)).

Our main comparative statics results are as follows. First, the wage is increasing in an index of prices $r(\mathbf{p}, \tau^s)$ and decreasing in coercion C . See equation (2). Second, coercion is increasing in the number of planters. See equation (5). Third, in the absence of coercion, wages are given by $w = r(\mathbf{p}, \tau^s)$ (equation (2)) so that we have a benchmark for competitive wages that deals explicitly with the crop substitution problem identified in Subsection 1.5.1. Fourth, planter strength N^* is decreasing in $\bar{W}/(1 - O)$ and $\bar{W}/(1 - O)$ has no direct impact on anything but N^* .³¹

²⁹ The classic example is the eighteenth-century Maroons operating in the mountainous interior of Jamaica.

³⁰ From equation (6), $\partial N^*/\partial \bar{W} = [(1 - O)\pi_N(N^*)]^{-1} < 0$ because $\pi_N(N^*) < 0$. Likewise, $\partial N^*/\partial O < 0$. While we do not need to sign $\partial^2 N^*/(\partial O \partial \bar{W})$, note that if $\pi(N^*)$ is linear in N in the neighbourhood of N^* then an increase in O increases $\partial N^*/\partial \bar{W} = [(1 - O)\pi_N(N^*)]^{-1}$, i.e., a higher probability of revolt makes N^* more sensitive to the returns from staying in England.

³¹ One could extend the model to include a market for plantation land. We have abstracted from this because land prices are not even remotely consistently available over time or across colonies.

3. The Evidence

Our main hypothesis is that the powerful Caribbean planter elite held enough sway over government to institute forms of legal coercion such as the incarceration of squatters, which were aimed at reducing wages. We begin by testing this hypothesis in OLS regressions of coercion (C_{it}) and wages (w_{it}) on the relative economic power of the planters (N_{it}):

$$w_{it} = \beta_N^w \cdot N_{it} + \gamma_X^w \cdot X_{it} + \lambda_i^w + \lambda_t^w + \epsilon_{it}^w; \quad (7)$$

$$C_{it} = \beta_N^C \cdot N_{it} + \gamma_X^C \cdot X_{it} + \lambda_i^C + \lambda_t^C + \epsilon_{it}^C, \quad (8)$$

where λ_i are colony fixed effects, λ_t are year fixed effects or year trends, and X_{it} are controls.

Data: Our wage data come from the *Blue Books*, which report wages for ‘praedial’ or agricultural workers. This was the wage paid to plantation workers. These wages were the largest component of the cost of the most important economic activity in the colonies (sugar). It is thus not surprising that wages were a constant subject of discussion in contemporary sources. The *Blue Book* wage data are identical to the sporadic data from reliable sources discussing wages for sugar cultivation (e.g., West India Royal Commission, 1897, p. 107). Further, the *Blue Books* themselves are sometimes the source of data quoted by contemporaries (e.g., Sewell, 1861). In short, the *Blue Books* reliably reported the well-known and very public data on wages for plantation work. Appendix B.1 discusses wages at greater length.^{32,33}

Our coercion data are incarceration rates per capita from the *Blue Books*. The *Books* report the daily average number of prisoners, averaged over the year, for 1838 to 1913. These data contain incarcerations for all reasons, but as discussed below and in Appendix B.3 our best estimate is that two-thirds of incarcerations at the start of our period were associated with legal coercion.³⁴ New incarcerations per capita (expressed as a percent) had a sample mean of 1.1%, indicating that 1.1% of the population entered jail each year. Incarceration rates are admittedly a fairly narrow measure of what was in reality a *bundle* of policies of legal coercion. The reason we focus on incarceration rates is that it is the only empirical counterpart to legal coercion that we could consistently code for the entirety of our wage sample. In the later years we are able to validate this measure using data on court sentencing that targeted smallholders. See Appendix B.3.

In equations (7)–(8), X_{it} are controls for observable factors that may have directly affected wages. Smallholder returns, and therefore plantations wages, were a function of exogenous price shocks and islands’ crop-specific soil productivity; see the wage equation (2). To construct $r_i(\mathbf{p}_t, \tau^s)$, we used the *Blue Books* to construct a 76-year panel of exports by colony and crop—generating a database containing exports by colony and year for 17 products accounting for 98% of exports—and combined it with fine-grained information on islands’ agro-climatic conditions to develop suitability indexes for the most important crops. We then estimated a Fréchet-based structural model of crop choice, as in Costinot *et al.* (2016), to recover estimates of the $r_i(\mathbf{p}_t, \tau^s)$. $r_i(\mathbf{p}_t, \tau^s)$ is an index of smallholder revenue based on exogenous crop suitability and it captures endogenous crop-switching in response to changes in the relative price of crops. It is an exogenous, model-based prediction. See Appendix B.2 for details.

³² We work with nominal wages. The *Blue Books* report that the major components of the cost of living were largely imported from Britain (clothing and many staples such as flour and rice) so that all 14 colonies shared a common cost of living. It follows that the cost of living deflator is absorbed in the year fixed effects used in our regressions.

³³ Over 90% of the wage data are for a daily pay period and do not involve in-kind payments. Nevertheless, in Online Appendix Table F.4 we show that our results are not sensitive to adjustments for pay periods or in-kind payments.

³⁴ Brizan (1984, p. 134) arrives at a similar number for Grenada, 1850–70.

Also included in X_{it} are the major labour supply shocks discussed in Subsection 1.5.2. For this, we calculated the island-specific cumulative stock of East Indian immigrant arrivals over time. It turns out that this flow of migrants was heavily right-skewed. It only really affected Guyana, Trinidad and to a lesser extent Jamaica. For the Panama Canal shock, there are no destination-specific estimates of islands out-migration so we measure the shock with a time dummy for the years of Canal construction (1881–9 under the French and 1908–13 under the Americans) divided by distance from island centroids to Panama. See Appendix B.6 for details.

We control for colony size using the log of population and the log of total export revenues. We control for each colony's time-varying British support using the number of times the colony was discussed in the British Parliament.³⁵

OLS results: Table 1 reports OLS estimates of equations (7) and (8), i.e., the effects of N_{it} on w_{it} and C_{it} . N_{it} is measured as sugar's share of exports. Columns 1–4 show results with w_{it} as the outcome. As a baseline, column 1 includes only a linear and quadratic time trend, and the Fréchet-based index of smallholder export prices $r_i(\mathbf{p}_i, \tau^s)$. Consistent with the wage equation (2) in the model, we see that smallholder export prices positively impact wages. In column 2 we add the controls for Indian immigration and the construction of the Panama Canal, i.e., the two most important labour supply shocks in the Caribbean during this period. As expected, Indian immigration reduced wages in the islands.³⁶ Proximity to the Panama Canal during the period of its construction had a positive but insignificant effect on wages. Column 3 additionally controls for factors that are likely endogenous: as expected, population growth lowered wages and higher economic activity raised wages. However, their inclusion does not affect the relationship between wages and N_{it} . The negative coefficient on the Hansard indicates that declines in parliamentary discussions of a colony correlate with declines in the colony's wage. This is useful because it rules out a potential source of reverse causality: as a colony's wage rose, sugar became unprofitable, Britain lost interest in the colony and stopped subsidising its white planter elite, and so N_{it} fell. That is, the Hansard result is not compatible with w_{it} causing N_{it} through a British subsidy channel.

In column 4 we use year fixed effects instead of the quadratic time trend. With year fixed effects the export price index is never close to significant. We therefore do not include it in any regressions with year fixed effects. This is done for expositional clarity and its exclusion has no bearing whatsoever on any of the other coefficients. Column 5 again adds the endogenous controls. Summarising columns 1–5, the partial correlation between changes N_{it} and wages is highly significant, very stable and appears economically large. The estimate $\hat{\beta}_N^w$ in column 4 says that in a colony like Grenada where sugar's share in exports had been reduced to zero by the end of the period, wages had increased by about 38% more over the 76 years than in a colony like Barbados where sugar's share in exports remained close to 1 at the end of the period.

The reader may worry about inference: we always cluster standard errors at the colony level, as this is intuitively appealing for our panel setting. However, with only 14 clusters we are naturally concerned about the asymptotic theory underlying standard clustering approaches (Moulton, 1986). For our core causal coefficients, we therefore also bootstrapped standard errors using the wild bootstrap, e.g., Cameron and Miller (2008); Davidson and MacKinnon (2010). These

³⁵ Data are from the parliamentary Hansard and are available at <https://hansard.parliament.uk/search>.

³⁶ The effect of immigration was statistically significant and economically large, but it only really affected two colonies. It depressed wages by 0.31 log points in Guyana ($-0.028 \times \ln(230,000)$) and by about 0.15 log points in Trinidad.

Table 1. OLS Effect of the Plantation System (N_{it}) on Wages (w_{it}) and Coercion (C_{it}).

	w_{it} : log wage					C_{it} : incarceration (per cap.)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Outcome										
N_{it} : Sugar exports as share of total exports	-0.3667 [0.0067]	-0.3633 [0.0078]	-0.3613 [0.0117]	-0.3831 [0.0080]	-0.3873 [0.0119]	0.4459 [0.0319]	0.4577 [0.0259]	0.4918 [0.0273]	0.4183 [0.0322]	0.4535 [0.0305]
Fréchet smallholder export price index $_{it}$	0.3256 [0.0027]	0.3263 [0.0032]	0.2578 [0.0273]			0.1608 [0.2844]	0.2149 [0.1556]	0.2657 [0.1442]		
log (net immigration) $_{it}$		-0.0266 [0.0023]	-0.0244 [0.0017]	-0.0224 [0.0110]	-0.0181 [0.0171]		-0.0013 [0.9326]	-0.0099 [0.6682]	0.0065 [0.7073]	-0.0009 [0.9736]
log(1/(dist to Panama)) $_i \times$ D(1881–9)(1908–13) $_t$		0.0040 [0.3442]	0.0036 [0.4232]	0.0265 [0.6997]	0.0338 [0.5961]		-0.0185 [0.0782]	-0.0184 [0.0867]	0.3637 [0.0571]	0.3756 [0.0493]
log(population) $_{it}$			-0.1197 [0.2866]		-0.1281 [0.1810]			0.1940 [0.3896]		0.1193 [0.6527]
log(value total exports) $_{it}$			0.0716 [0.0629]		0.0501 [0.2569]			-0.0202 [0.8468]		0.0235 [0.8140]
log Hansard-mentions $_{it}$			0.0019 [0.7091]					0.0076 [0.6692]		
Time controls	$t + t^2$	$t + t^2$	$t + t^2$	t-fe	t-fe	$t + t^2$	$t + t^2$	$t + t^2$	t-fe	t-fe
Observations	908	908	908	908	908	798	798	798	798	798
R^2	0.711	0.724	0.728	0.768	0.770	0.511	0.516	0.518	0.572	0.573

Notes: Column 1 includes a quadratic time trend and the Fréchet-based index of smallholder export prices $r_i(\mathbf{p}, \tau^s)$. Column 2 adds the most important labour supply shocks in the Caribbean during this period. Column 3 additionally controls for the log of population (a proxy for size), the log of total exports (a proxy for economic size), and parliamentary Hansard mentions (a proxy for British interest in the colony). Column 4 replaces the quadratic time trend with year fixed effects, and drops the export price index. Column 5 adds the same controls as column 3. In columns 6–10, we repeat the same specifications for incarceration as the outcome. Standard errors are clustered by colony and the corresponding p -values are in square brackets.

Table 2. *OLS Effect of Sugar Plantations on Wages (w_{it}) and Coercion (C_{it}).*

Outcome	w_{it} : log wage					C_{it} : incarceration (per cap.)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
N_{it} : plantation exports as share of total exports	-0.5076 [0.0030]	-0.5131 [0.0026]	-0.5084 [0.0033]	-0.5504 [0.0028]	-0.5456 [0.0040]	0.6923 [0.0140]	0.6971 [0.0148]	0.6963 [0.0148]	0.6451 [0.0185]	0.6502 [0.0182]
Time controls	$t + t^2$	$t + t^2$	$t + t^2$	t-fe	t-fe	$t + t^2$	$t + t^2$	$t + t^2$	t-fe	t-fe
Observations	908	908	908	908	908	798	798	798	798	798
R^2	0.717	0.730	0.734	0.776	0.778	0.518	0.522	0.523	0.578	0.578

Notes: This table replicates Table 1 with N_{it} instead measured as plantation crops' export share.

were published in earlier working paper versions of the paper, and never pushed the statistical significance of any of our coefficients of interest below 10%.

In columns 6–10, we repeat these specifications for C_{it} as the outcome. The resulting estimates suggest that a complete collapse of the plantation system (from 1 to 0) is associated with a decrease in incarceration rates of about one person per two-hundred in a year, roughly one half the mean incarceration rate in the data.³⁷ Consistent with the model, the labour supply (wage) controls have no direct effect on coercion. Only the Panama Canal control is significant, but switches signs across specifications. Our model, our intuition and our regression results all tell us that labour supply shocks do not effect incarceration.

In the remainder of the paper, we estimate all results with two core specifications. The first corresponds to columns 2 and 7. The second corresponds to columns 4 and 9. That is, we include either a quadratic time trend or year fixed effects, but omit the potentially endogenous controls for population, total exports and the Hansard. (Including these endogenous controls alters none of the results.)

In Table 2 we report the corresponding coefficients on N_{it} where N_{it} is now measured as plantation crops' export share. The results are similarly robust and stable in magnitude. Across the board, the coefficients in Panel B are about one-third larger than in Panel A, but the economic effect is almost the same because sugar's share in exports dropped less than the share of all plantation crops combined. The estimate $\hat{\beta}_N^w$ in column 4 says that in a colony like Grenada where plantation crops' share in exports had been reduced to about one-quarter by the end of the period, wages had increased by about 38% (0.55×0.75) more over the 76 years than in a colony like Barbados where the plantation system continued to completely dominate.

Endogeneity of N_{it} aside, the reader may have a number of concerns with Tables 1 and 2. One potential concern pertains to the claim (and in the model, our assumption) that agricultural wages are only paid in the plantation sector. We validate this claim in Appendix B.1 where we provide prima facie evidence that wage employment was the domain of the plantation system, and that the only common alternative for plantation workers was independent smallhold farming, as opposed to wage labour outside of agriculture.

Another potential concern pertains to the accuracy of the reported wage data. Wages sometimes included an in-kind component that usually involved a cottage and a small plot. For two-thirds of our sample the *Blue Books* indicate whether the wage includes in-kind payments. Only 9% of these observations include in-kind payments. Nevertheless, Online Appendix Table F.4 shows that our results are robust to an adjustment for in-kind payments. Also, wages are almost always reported as daily, but for 5% of our observations data are reported as weekly or monthly. When this happens, the *Blue Books* explicitly indicate that the data are for five-day weeks or 20-day months and so are easily converted to daily wages. Nevertheless, Online Appendix Table F.4 shows that

³⁷ To put this number in context, in the USA in 2016, the stock of incarcerated was 0.7 persons per 100.

our results are robust to adjustments for alternative conversions of weekly and monthly wages into daily wages.

Another potential concern pertains to using incarceration rates as our measure of legal coercion. Mitigating this concern is the fact that the historical literature is quite clear that incarceration was indeed often the result of legal coercion aimed at smallholders. In Appendix B.3 we document this by very detailed category of offence for the Leewards for the start of our period. For the end of our period (1871–1913) we have court sentences but only by broad categories of offences, and only one of these—offences against property—maps into the legal coercion discussed in Subsection 1.1. In Appendix B.3 we show that this category's share of total court sentences correlated positively with both N_{it} and C_{it} .

Identification: The OLS estimates reported thus far should be interpreted with caution because N_{it} is likely to be endogenous in both the wage and coercion equations. In the wage equation, for example, productivity growth in plantation agriculture would have increased both N_{it} and w_{it} . Unobserved differential productivity growth would thus bias the OLS results against finding a negative effect of N_{it} on w_{it} . In the coercion equation, for another example, if, unobserved to the econometrician, peasants in some but not all colonies were actively resisting the plantation system this could have evoked a planter response (increased C_{it}) while also undermining the plantation system (decreased N_{it}).³⁸ This would bias the OLS results against finding a positive effect of N_{it} on C_{it} .

Equation (6) offers a path towards an instrument for N_{it} . A British investor chooses between investing in the Caribbean, which returns $(1 - O_i)\pi(N_{it})$, or investing elsewhere, which returns $\bar{W}_t$. Across colonies, those colonies exogenously endowed with more hinterland O_i are less likely to receive the investment. Over time, improvements in non-Caribbean investment opportunities $\bar{W}_t$ make it less likely that any colony receives the investment. Thus, the larger is $O_i \cdot \bar{W}_t$ the smaller is N_{it} .³⁹ We already have a hinterland measure O_i . We also argued in Subsection 1.4 that British exports to non-Caribbean countries is a good measure of non-Caribbean opportunities $\bar{W}_t$.⁴⁰ We expect that a higher $O_i \cdot \bar{W}_t$ leads to a lower N_{it} , i.e., $O_i \cdot \bar{W}_t$ has a negative sign in the first stage.

The Virgin Islands' experience—where the plantation system collapsed because of two major hurricanes despite low values of O_i —suggests the need for a second, hurricane-based instrument HDI_{it} for N_{it} . (See Subsection 1.5.3.) It is important to be clear about what role HDI_{it} plays: hurricanes are not central to our paper but we need them to fit the variation in N_{it} for the Virgin Islands. Without the Virgin Islands, hurricanes are no longer needed to fit N_{it} , but we prefer to retain them in order to reflect the full universe of British Caribbean sugar colonies, and because they were a particularly striking example of the mechanisms we highlight.

First Stage and Reduced Form Estimation: In moving on to our TSLS estimation, we propose the following first stage equation

$$\text{First Stage: } N_{it} = \beta_O^N \cdot O_{it} + \beta_H^N \cdot HDI_{it} + \gamma_X^N \cdot X_{it} + \lambda_i^N + \lambda_t^N + \epsilon_{it}^N, \quad (9)$$

as well as the associated reduced form relations

³⁸ For instance, Caribbean historiography suggests that the unmeasured proliferation and influence of local missionaries was an important factor that encouraged civil disobedience and undermined the planters; see for example Dookhan (1977, p. 156), Lewis (1986, ch. 3), McLewin (1987, pp. 85–7) and Holt (1992, ch. 7).

³⁹ In the model $O_i \cdot \bar{W}_t$ appears in equation (6), though parameterised slightly differently.

⁴⁰ We include both the British and non-British Caribbean in order to net out Cuba, which produced about one-third of world sugar cane in our period.

$$\text{Reduced Form (Wages): } w_{it} = \beta_O^w \cdot O_{it} + \beta_H^w \cdot HDI_{it} + \gamma_X^w \cdot X_{it} + \lambda_i^w + \lambda_t^w + \epsilon_{it}^w; \quad (10)$$

$$\text{Reduced Form (Coercion): } C_{it} = \beta_O^C \cdot O_{it} + \beta_H^C \cdot HDI_{it} + \gamma_X^C \cdot X_{it} + \lambda_i^C + \lambda_t^C + \epsilon_{it}^C, \quad (11)$$

where i indexes colonies and t indexes years, the λ s are fixed effects, and X_{it} are the same control variables as in the OLS. Columns 1–4 of Table 3 report estimates of the *First Stage* equation (9) for our two favoured specifications and for the two different measures of planter power. In columns 1–4, $\hat{\beta}_O^N$, the coefficient on $O_i \cdot \bar{W}_t$ indicates that in islands with large hinterlands the plantation system declined faster in response to improving non-Caribbean opportunities for British interests. The hurricane damage index HDI_{it} has a powerful negative impact on the plantation system. In columns 5–8, we report estimates of the two reduced-form equations (10) and (11). The estimate $\hat{\beta}_O^w$ in column 3 says that compared to Barbados, wages in Grenada (where half the land was hinterlands) rose substantially more in response to an increase in $\bar{W}_t$. At the same time, columns 7–8 show a significant effect on incarceration rates working in the opposite direction.

Clearly, the time-varying component of $O_{it} = O_i \cdot \bar{W}_t$ is a strongly trending variable; see Panel (b) of Figure 3. Therefore, a potential concern with the results reported in Table 3 is whether the time path of the effect of O_i on the plantation system and on wages actually matches the evolution of $\bar{W}_t$. The most flexible way to ask this question is to re-estimate equations (9) and (10), replacing O_{it} with an interaction between O_i and year fixed effects (while continuing to include year fixed effects on their own as in columns 2, 4, 6 and 8 of Table 3).

Figure 4 plots the coefficients estimates on this interaction that result from doing so, with 1838 omitted because it gets absorbed by the colony fixed effects as the first year. The solid thick line shows a clearly discernible negative effect of O_i on N_{it} , and this effect sets in around the mid-1860s. The solid thin line shows a positive effect of O_i on wages kicking in around the same time.⁴¹ We view these time paths as consistent with the time path of $\bar{W}_t$, and thus with our instrument: it is natural that the interaction effect on N_{it} in Figure 4 could not drop below -1 since N_{it} is bounded below at zero; and it is therefore equally natural that the positive effect of O_i on w_{it} flattens off.

TSLS Estimation: We now turn to the causal effect of the declining strength of the plantation system on coercion and wages, estimating equations (7) and (8) with TSLS, and using equation (9) as the first stage for both. Table 4 reports the results for our two preferred specifications. Columns 1–4 report estimates of equations (7) and columns 5–8 report estimates of equation (8). Columns 1–2 and 5–6 report results when N_{it} is measured as sugar's share of exports. Columns 3–4 and 7–8 report results when N_{it} is measured as plantation crops' share of exports. The estimates $\hat{\beta}_N^w$ and $\hat{\beta}_N^C$ are very robust in magnitude within pairs of columns (1–2, 3–4, etc.), indicating results do not hinge on the particular specification of the time controls. The F -statistics on the First Stage instruments are comfortably above critical threshold levels. The TSLS magnitudes $\hat{\beta}_N^w$ in columns 1–2 and 3–4 are about 60% larger than the OLS estimates reported in columns 2 and 4 of Tables 1 and 2. The TSLS magnitudes $\hat{\beta}_N^C$ in columns 5–6 and 7–8 are around twice the OLS estimates reported in columns 6 and 8 of Tables 1 and 2. In combination, this suggests that the OLS estimates are downward biased, consistent with the discussion at the start of the identification section.

⁴¹ There was also a temporary spike in wages associated with a cholera epidemic that spread through the Caribbean in the early 1850s.

Table 3. *First Stage and Reduced Form.*

Outcome	N_{it} : Sugar's export share		N_{it} : Plantations' export share		w_{it} : log wage		C_{it} : Incarceration (per cap.)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$O_t \times$ English exports to all non-Caribbean, HDI_{it}	-0.4819 [0.0000] -0.0680 [0.0000]	-0.5238 [0.0001] -0.0751 [0.0000]	-0.2365 [0.0012] -0.0625 [0.0000]	-0.2611 [0.0030] -0.0646 [0.0000]	0.2351 [0.0153] 0.0483 [0.0001]	0.2724 [0.0075] 0.0565 [0.0001]	-0.4577 [0.0056] -0.0857 [0.0000]	-0.4121 [0.0287] -0.0775 [0.0094]
Time controls	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe
Observations	1,018	1,018	1,018	1,018	908	908	798	798
R ²	0.853	0.867	0.843	0.851	0.721	0.772	0.526	0.580

Notes: Columns 1–4 present estimates of equation (9), the *First Stage* relation between the instruments and the two measures of N_{it} . Columns 5–8 present estimates of equations (10) and (11), the *Reduced Form* relations between the instruments and the two main outcomes w_{it} and C_{it} . For each outcome, we present the results for our two preferred specifications which include a quadratic time trend (odd-numbered columns) or year fixed effects (even-numbered columns). These correspond to columns 2 and 4 of Table 1. For brevity, we do not report coefficients on any of the controls that appear in Table 1. All specifications include colony fixed effects. Standard errors are clustered by colony, p -values are in square brackets.

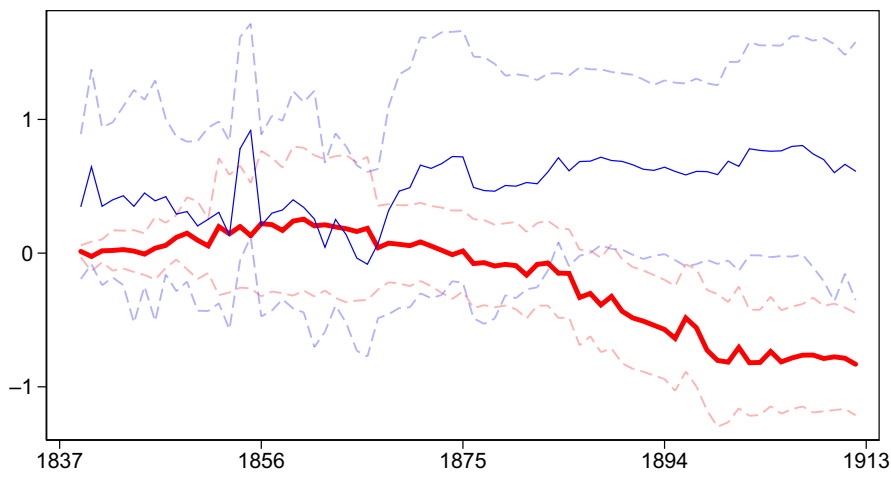


Fig. 4. *The Year-Specific Effect of O_i on N_{it} and on w_{it} .*

Notes: This figure reports the point estimates and 95th-percentile confidence bands on O_{it} in two estimations of regression equations (9) and (10), when O_{it} is replaced by a flexible interaction of O_i with year fixed effects (while also including year fixed effects on their own). The solid thick line reports on the estimated effect on N_{it} in (9), the solid thin line reports on the effect on w_{it} in (10); dashed lines are confidence bands.

Table 4. *TSLS Effect of the Plantation System (N_{it}) on Wages (w_{it}) and Coercion (C_{it}).*

Outcome	w_{it} : log wage				C_{it} : incarceration (per cap.)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
N_{it} (share of exports)	-0.5789 [0.0006]	-0.5811 [0.0084]	-0.9091 [0.0000]	-0.9641 [0.0018]	1.0675 [0.0001]	0.9018 [0.0057]	1.6282 [0.0000]	1.4638 [0.0003]
Fréchet smallholder export price index $_{it}$	0.3380 [0.0001]		0.3161 [0.0004]		0.1636 [0.2146]		0.2050 [0.1395]	
log (net immigration) $_{it}$	-0.0260 [0.0081]	-0.0219 [0.0561]	-0.0283 [0.0001]	-0.0241 [0.0226]	0.0011 [0.9686]	0.0083 [0.7729]	0.0062 [0.7961]	0.0125 [0.6369]
log(1/[dist to Panama]) $_i \times$ D(1881–9) [(1908–13)] $_t$	0.0046 [0.2249]	0.0533 [0.3038]	0.0038 [0.3040]	0.0596 [0.2997]	-0.0201 [0.0513]	0.2946 [0.1394]	-0.0181 [0.0755]	0.3246 [0.1383]
N_{it} measured based on	Sugar		Plantations		Sugar		Plantations	
Time controls	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe
Observations	908	908	908	908	798	798	798	798
F-statistic (instruments)	22.9	21.5	24.8	22.3	20.2	21.6	19.2	21.1

Notes: This table reports the TSLS specifications corresponding to the specification in Tables 1 and 2. Columns 1–4 report the estimates of the wage equation (7). Columns 5–8 report the estimates of coercion equation (8). In columns 1–2 and 5–6, N_{it} is measured as sugar’s share of exports and plantation crops’ share of exports, respectively. For each outcome, we present the results for our two preferred specifications which include a quadratic trend or year fixed effects. This corresponds to columns 3–4 of Table 1. For expositional clarity, we do not report coefficients on any of the controls. All specifications include colony fixed effects. Standard errors are clustered by colony.

While we argue that the agro-climactic and international factors that underlie our instrument $O_i \cdot \bar{W}_t$ have no direct impact on Caribbean wages or coercion, $O_i \cdot \bar{W}_t$ may nevertheless fail to satisfy the exclusion restriction because it is correlated with omitted variables that influence wages or coercion. In this section we present a range of specifications which address this concern.

Given the presence of both colony and year fixed effects, the precise concern is that $O_i \cdot \bar{W}_t$ is correlated with differential trends across colonies. For concreteness consider the following example, chosen because it is particularly damaging to our exclusion restriction. (1) Suppose that in some colonies labour supply trended up, leading to a fall in wages and a resulting expansion of sugar ($w_{it} \downarrow \rightarrow N_{it} \uparrow$) while in other colonies labour supply trended down, leading to a rise in wages

and a resulting contraction of sugar ($w_{it} \uparrow \rightarrow N_{it} \downarrow$). (2) Suppose that we have not controlled for these differential trends in labour supply. (3) Suppose that our instrument is correlated with these differential trends. Under (1)–(3), our exclusion restriction is violated and a TSLS regression of w_{it} on N_{it} would be more negative than the true coefficient. Thus, differential labour supply trends can potentially explain away our results. In Online Appendix G.1–G.5, we show that controlling for a variety of different trends leaves the point estimates and significance levels of our results qualitatively unchanged.

4. Extensions

In this section, we focus on showing that the negative effect of planter power on wages was in fact explained by the positive effect of planter power on coercion; and that coercion in our setting was of the type described in Subsections 1.1–1.2.

The history, theory and econometrics presented thus far all point towards the following causal relations: $O_{it} \downarrow \rightarrow N_{it} \downarrow \rightarrow C_{it} \downarrow \rightarrow w_{it} \uparrow$. However, we have not shown that there is a causal impact of C_{it} on w_{it} ; we have only shown that N_{it} impacts C_{it} and w_{it} . To remedy this, we estimate a model that regresses w_{it} on C_{it} , and instruments C_{it} with O_{it} ; reported in Online Appendix H (*Extension I*). The model establishes a causal impact of coercion on wages ($C_{it} \rightarrow w_{it}$). This, however, still does not quantify the relative importance of the politics–coercion mechanism we postulate ($N_{it} \rightarrow C_{it} \rightarrow w_{it}$) because our estimates of the impact of plantation power on wages ($N_{it} \rightarrow w_{it}$) reported in Section 3 may be capturing our postulated mechanism *along with* other political mechanisms through which N_{it} impacts w_{it} . It is therefore of interest to estimate how much of the impact of N_{it} on w_{it} operates via coercion as opposed to other political mechanisms.

A decomposition like this is done through a ‘mediation analysis’, which asks to what extent the ‘total effect’ of a treatment variable (e.g. N_{it}) on a ‘final outcome’ (e.g. w_{it}) is explained by an intermediate outcome or mechanism (e.g., C_{it}) that is also an outcome of N_{it} . See for example Imai *et al.* (2010; 2011); Heckman *et al.* (2013); Heckman and Pinto (2015). We use the methodology developed in Pinto *et al.* (2019) for performing mediation analysis in an IV setting like ours, estimate that upwards of 75% of the estimated effect of N_{it} on w_{it} operates through coercion C_{it} ; reported in Online Appendix H (*Extension II*).⁴²

One aspect of legal coercion that is emphasised in the historical discussion in Subsection 1.2 is the distinction between officially sanctioned *de jure* power and unofficially sanctioned *de facto* power. We now supplement this historical discussion with econometric evidence by investigating the consequences of the Caribbean Incumbered Estates Act (IEA) of 1854, an exogenous shock to planters’ *de facto* relative to their *de jure* power in the exercise of coercion. The IEA originated in the Irish Potato Famine (1845–9), which left Irish estate owners without operating capital and deeply in debt. In response, the Irish Incumbered Estates Act of 1849 strengthened creditor rights by allowing forced foreclosures (estate sales), which allowed capital to return to the Irish estates. The success of this legislative innovation, the result of a famine that was exogenous to the British West Indies, led the British Parliament to enact a Caribbean IEA in 1854.⁴³

⁴² Results reported in Online Appendix Table H.9.

⁴³ Before 1838, in the heyday of sugar, many plantations accumulated large encumbrances, that is, financial commitments to pay annual stipends to family members and to repay loans incurred for capital projects such as mills. After 1838, rising wages and falling sugar prices led many planters to ignore their encumbrances. They could do so with impunity because the law subordinated creditors to both the plantation owners and to the London-based merchant houses who

The most important consequence of the Caribbean IEA was a change in the composition and identity of the planter elite, because buyers of encumbered estates were often London-based merchant houses rather than local Caribbean planters (Hall, 2011, p. 97). This meant that planters with deep local roots at the parish level were being replaced by London-based merchant houses that set up offices in colonial capitals, which changed the way legal coercion operated. The *de facto* power of the traditional planters elite over local parish police and magistrates made incarceration an easy method of intimidation and coercion. By contrast, the new merchant-house planters that were based in the islands' metropolises had more direct access to influence the colonial administration and the islands' legislatures (*de jure* power). We therefore conjecture that the IEA led coercion to become more legislated, i.e., *de jure* based. To test this we need a measure of legislated coercion. Planter elites enacted many coercive laws, such as discriminatory taxes on smallholds and licensing requirements for the transportation and sale of smallhold crops. Most of the coercive laws are very difficult to code up as a regressor.⁴⁴ A less complex coercive law is the tariff on imports of foodstuffs. As noted in Subsection 1.1, food imports competed with smallhold crops so low foodstuff tariffs made smallholding less attractive. Low foodstuff tariffs were *de jure* coercive.

We also need to code up the IEA as a regressor. Using the full records of IEA court proceedings during the period it was in effect (1858–89), we identified all 398 cases of court-ordered sales of plantations.⁴⁵ Our reading of the IEA court cases is that the petitions for forced sale were often initiated by family members, sometimes distant family members, who had randomly fallen on hard times. To be safe, we also instrument court-ordered sales with court petitions. There were 523 IEA petitions in total, of which 398 led to court-ordered sales. Figure 5 shows the cumulative sales and petitions by colony over time.⁴⁶

To summarise, we will regress wages, incarceration rates and foodstuff tariffs on IEA sales and instrument IEA sales with IEA petitions. We will also include N_{it} as before. Panel B of Table 5 reports first stages. There is a very clean statistical separation between the instruments for IEA sales and N_{it} . That is, IEA petitions have a strong impact on IEA sales, but no effect on N_{it} (see columns 1–4). Likewise, O_{it} and HDI_{it} have a strong effect on N_{it} , but no effect on IEA sales (see columns 5–6).

Panel A of Table 5 reports the TSLS estimates of the effects of N_{it} and IEA sales on wages, incarceration rates and foodstuff tariffs. To compare magnitudes, we have scaled IEA sales by their mean.⁴⁷ The coefficients on IEA sales are thus the average effect of the IEA. The effects of N_{it} on wages and incarceration rates (columns 1–4) are almost identical to our baseline Table 4 results. The average total impact of the IEA on wages was to lower them by approximately 5.89 percent (in column 2). In contrast, N_{it} declined on average by 0.5 and so lowered wages

profited from selling sugar. This and the subsequent discussion is based on Cust (1859, pp. 8–15). Cust was secretary to the West Indian Incumbered Estates Commission. See also Hall (2011, pp. 94–7).

⁴⁴ These laws appear in the *Blue Books*. For example, most islands had regressive land and property taxes, but some taxes were levied in values, others were levied on acreage, and yet others had ad hoc qualifications such as tax adjustments for windmills, which were aimed at relieving planters.

⁴⁵ Hall (2011, p. 96) gives a slightly lower number of 382. He likely pulled this number from the 1884 Royal Commission Report on the IEA, which preceded the actual ending of the IEA by five years (Crossman and Baden-Powell, 1884). To the best of our knowledge we are the first to code up the IEA records since 1884. The sales and petitions were recorded in the Colonial Office Records Series CO 318-282-50 and CO-441-2-11, respectively.

⁴⁶ The IEA was passed in London but needed to be incorporated on each island to apply there. St Lucia, Guyana, Trinidad and Barbados never incorporated the IEA, and thus no IEA sales took place in those colonies (Hall, 2011, p. 97).

⁴⁷ In the full panel data, the average observation (over time and across all islands with any sales) had a cumulative number of IEA sales of 28. See Figure 5.

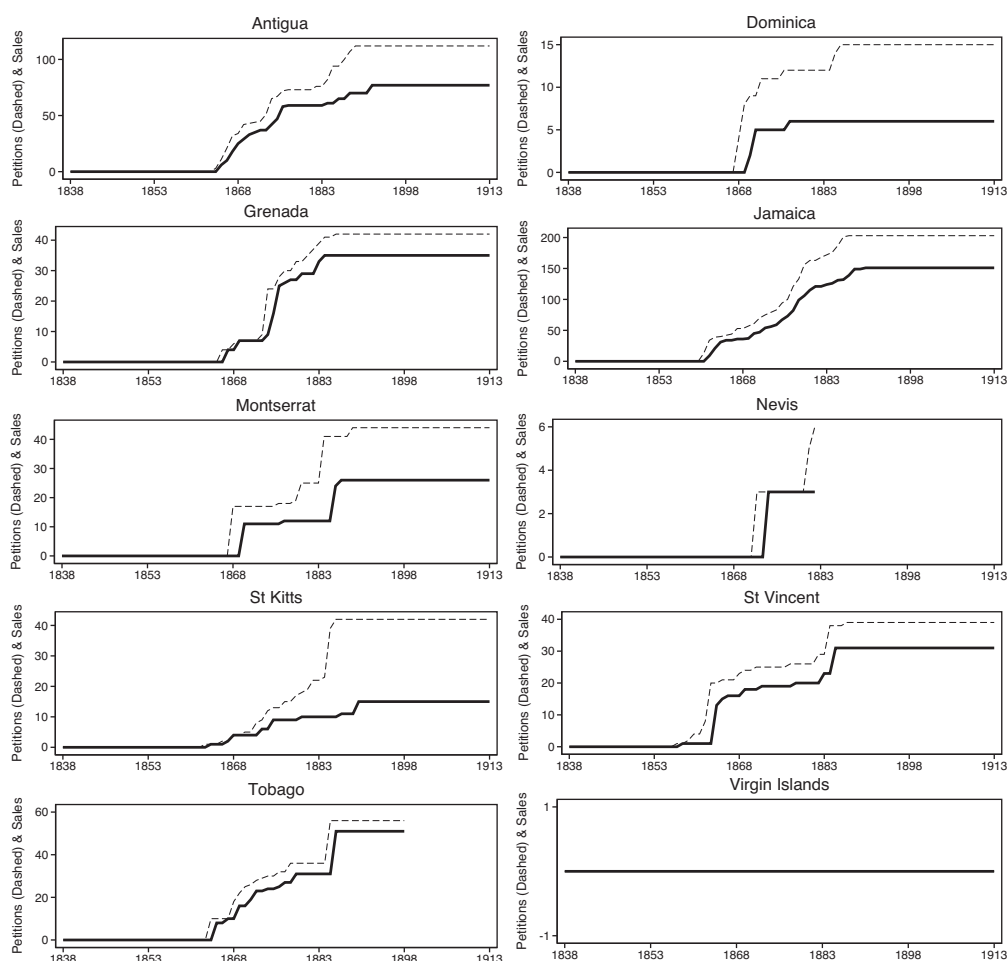


Fig. 5. *Incumbered Estate Act Petitions and Sales.*

Notes: This figure shows colony-specific time series of the cumulative number of IEA petitions and sales. In total, there were 523 IEA petitions (by claimants) to force sales and 398 plantations were actually sold through the IEA between 1858 and 1889. The IEA was passed in London but needed to be incorporated on each island to apply there. St Lucia, Guyana, Trinidad and Barbados are not in the figure because they never incorporated the IEA, and thus no IEA sales took place in those colonies (Hall, 2011, p. 97).

by approximately 28.4 percent ($=0.5 \times 0.568$). This suggests that the collapse of the plantation system (N_{it}) was the much more fundamental shock to labour markets than the introduction of the IEA, which is fully consistent with the historical literature. This eyeball observation is also borne out in a statistical test of the equality of the two coefficients of interest (on N_{it} and on IEA sales), which we report at the bottom of Panel A. When it comes to incarceration, the effect of IEA sales was not only much smaller than that of N_{it} but was in fact statistically not distinguishable from zero (in columns 3–4). This stands in contrast to the effects on foodstuff tariffs, where columns 5–6 show that as the plantation system weakened (N_{it} fell), foodstuff tariffs rose to the benefit of smallholders, but that this effect is statistically insignificant. In contrast, and as hypothesised,

Table 5. *The Effect of the IEA.*Panel A. *TSLS estimation of effects of N_{it} and IEA sales $_{it}$ on three outcomes*

Outcome	w_{it} : log wage		C_{it} : incarceration (per cap.)		Import duties flour (£)	
	(1)	(2)	(3)	(4)	(5)	(6)
N_{it} : sugar exports	-0.7151	-0.5683	1.2239	0.9346	-1.0126	-0.6124
share of total exports	[0.0004]	[0.0181]	[0.0001]	[0.0031]	[0.3102]	[0.5148]
IEA sales $_{it}$	-0.0388	-0.0589	-0.0662	-0.0547	-0.3593	-0.4226
	[0.0495]	[0.0258]	[0.2909]	[0.4050]	[0.0026]	[0.0298]
Time controls	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe
Observations	908	908	798	798	942	942
p-value: Test[2 coefficients equal]	0.001	0.031	0.000	0.002	0.515	0.835
F-statistic (instruments)	18.878	15.111	17.581	17.279	21.146	17.397

Panel B. *First stage for two endogenous regressors: N_{it} and IEA sales $_{it}$*

Outcome	N_{it} : plantations' export share		N_{it} : sugar's export share		IEA sales $_{it}$	
	(1)	(2)	(3)	(4)	(5)	(6)
$O_i \times$ English exports	-0.2399	-0.2359	-0.4452	-0.4925	0.0517	0.1095
to all non-Caribbean $_t$	[0.0075]	[0.0025]	[0.0001]	[0.0001]	[0.5780]	[0.3414]
HDI $_{it}$	-0.0665	-0.0665	-0.0677	-0.0718	0.0076	0.0140
	[0.0000]	[0.0000]	[0.0000]	[0.0000]	[0.3717]	[0.2126]
Incumbered Estate Act	-0.0227	-0.0214	-0.0133	-0.0177	1.0367	1.0522
petitions $_{it}$	[0.1734]	[0.2438]	[0.5498]	[0.4786]	[0.0000]	[0.0000]
Time controls	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe
Observations	1,018	1,018	1,018	1,018	1,018	1,018
R ²	0.882	0.884	0.891	0.897	0.991	0.990

Notes: (a) Panel A presents the TSLS estimation of the effect on three outcomes. Online Appendix Table H.10 reports additional robustness checks on Panel A. Panel B presents the first stage results for two endogenous regressors, i.e., N_{it} and IEA sales $_{it}$. Because there are three different numbers of observations in Panel A, we report the first stage in Panel B for all available data. In Online Appendix Table H.11, we also report the first stage for each of the three sets of observations in Panel A. This does not materially affect any coefficients. (b) For each outcome, we present the results for our two preferred specifications, i.e., the second and third specifications shown in all results in Section 3. (c) Standard errors are clustered by colony, p-values in square brackets.

columns 5–6 also show the higher *de jure* access of the new merchant-planters brought in by the IEA to the islands' lawmakers led to a much more precisely estimated reduction in foodstuff tariffs.⁴⁸

In summary, the IEA sales provide evidence on the use of *de facto* versus *de jure* coercion. IEA sales did not increase *de facto* coercion as measured by incarceration rates (the results in columns 3–4 are insignificant), did increase *de jure* coercion as measured by lower foodstuff tariffs. The shape of legal coercion in the British West Indies varied with changes in the political power of the planter elite.

5. Conclusion

In this paper, we provide the first rigorous empirical evidence on the importance of legal coercion in informal developing-country labour markets. Following Lewis's famous assessment quoted

⁴⁸ Seeing as the point estimates of the effect of N_{it} on foodstuff tariffs are imprecise but large, we do not have the statistical power to reject that the effects of N_{it} and IEA sales $_{it}$ on tariffs are equal. In Online Appendix Table H.10 we report additional specifications where we control for general changes in public finance (revenues and expenditures) in our data.

at the beginning of this paper, we hypothesise that legal coercion aimed at depressing workers' outside options in the informal sector is an important factor in determining wages, and that the formal sector's influence over government is a key driver of the use of legal coercion.

We focus on a uniquely relevant empirical setting, studying 14 Caribbean plantation islands at the inception of free labour markets, i.e., starting right after the emancipation of slaves. We measure the formal sector's influence over government N_{it} by the plantation sector's share of overall production. To gain identification, we instrument for N_{it} with a interaction between the planters' time-varying reservation wages outside the Caribbean plantation system and variation across islands in the peasants' ability to evade the plantation system.

Controlling for possibly confounding crop price variation, labour demand and labour supply shocks, we find strong support for both hypotheses in the data: the plantation system had powerful effects on wages and legal coercion (see Tables 1 and 4). A complete collapse of the plantation system raised agricultural wages by 100% and lowered coercion by two times its mean level in the data. At the sub-island parish level we can also show that it lowered mortality rates. Pushing further towards identifying the causal mechanism linking these patterns, we find that upwards of three-quarters of the plantation system's effect on wages is explained by the plantation system's effect on coercion (see Online Appendix Table H.9). We also provide suggestive evidence that planters shaped legal coercion both through legislation passed in the islands as well as through personal connections to the police and judicial apparatus in the countryside.

In summary, Lewis was right.

Appendix A. Mathematical Proofs

Proof of Equation (1): Drop j superscripts, i and t subscripts, and $\mathbf{p}_t$ arguments. Define $\hat{T}_g \equiv T_g(\tau_g p_g)^\theta$. Let $r_g(\omega) = \hat{T}_g z_g(\omega)^\theta$ be revenue per plot generated by crop g . Then $r_g(\omega)$ has cumulative distribution $e^{-\hat{T}_g r_g^{-\theta}}$, density $\theta \hat{T}_g e^{-\hat{T}_g r_g^{-\theta}}$ and mean $\hat{T}_g^{1/\theta} \Gamma(1 - 1/\theta)$. As is well known from Eaton and Kortum (2002), $Pr\{r_k(\omega) < r \forall k \neq g\} = e^{-(\hat{T} - \hat{T}_g)r^{-\theta}}$ where $\hat{T} = \sum_{k=1}^G \hat{T}_k$ and $Pr\{\text{choose } g\} = \hat{T}_g/\hat{T}$. Hence, the joint probability of g being the optimal crop choice and $r_g(\omega) = r$ is

$$Pr\{\{\text{choose } g\} \text{ and } \{r_g(\omega) = r\}\} = Pr\{r_k(\omega) < r \forall k \neq g\} Pr\{r_g(\omega) = r\} = (\theta \hat{T} e^{-\hat{T} r^{-\theta}})(\hat{T}_g/\hat{T}).$$

The first term in parentheses is a Fréchet density with scale parameter $\hat{T}$ and hence with mean $\hat{T}^{1/\theta} \Gamma$. Hence the expected revenues generated by crop g are

$$r_g(\hat{T}_g) \equiv \mathbf{E}[r_g(\omega)] = \hat{T}^{1/\theta} \Gamma (\hat{T}_g/\hat{T}) = \hat{T}_g \hat{T}^{\frac{1}{\theta}-1} \Gamma. \quad (\text{A1})$$

Further, the expected revenues generated by all crops are

$$r(\hat{T}_g) \equiv \sum_g r_g(\hat{T}_g) = \hat{T}^{\frac{1}{\theta}} \Gamma, \quad (\text{A2})$$

where $\hat{T} \equiv \sum_{g=1} \hat{T}_g$. Substituting $\hat{T}_g \equiv T_g(\tau_g p_g)^\theta$ into this mean yields equation (1). $\square$

Characterisation of $C^*(N)$: Substituting equations (2)–(3) into (4) yields $W(C) = (\alpha l(N)N - L)C - \alpha C^\gamma + \alpha l(N)N[r(\mathbf{p}, \tau^p) - r(\mathbf{p}, \tau^s)] + Lr(\mathbf{p}, \tau^s)$, which is concave in C for $\gamma > 1$. Hence C^* is unique. $\partial W/\partial C = 0$ implies $(C^*)^{\gamma-1} = [\alpha l(N)N - L]/[\alpha \gamma]$. From the definition of $\bar{N}$ before

equation (5), $\alpha l(N)N - L > 0 \Leftrightarrow N > \bar{N}$. Hence the constrained ($C \geq 0$) optimal solution is $C^* = 0$ for $N < \bar{N}$ and equation (5) for $N \geq \bar{N}$.

Existence, Interior Solutions and Stability of N^* : The equilibrium number of planters N^* is given by equation (6). We first derive a sufficient condition which ensures an interior solution $N^* \in (0, L)$. Start by defining $\Delta r \equiv r(\mathbf{p}, \tau^p) - r(\mathbf{p}, \tau^s)$. A sufficient condition for an interior N^* is $\pi(0) > \bar{W}/(1 - O) > \pi(L)$. From equation (5), $C^*(0) = 0$ so that from equation (3), $\pi(0) = l(0)\Delta r$. Also, it is tedious but straightforward to show that $\pi(L) < \Delta r L^{1/(\gamma-1)}$. Hence a sufficient condition on parameters is $l(0)\Delta r > \bar{W}/(1 - O) > \Delta r L^{1/(\gamma-1)}$. To see stability, consider a plot of $\pi(N)$ and $\bar{W}/(1 - O)$ against N . As we move from $N = 0$ to $N = L$, $\pi(N)$ must cut the horizontal $\bar{W}/(1 - O)$ line from above, which is the requirement for stability. Finally, at such an intersection $N = N^*$ and $\pi_N(N^*) < 0$.

Appendix B. Data

We made use of two types of original data sources. The first is a new data set of geographic and agro-climatic conditions in the Caribbean. To calculate predetermined outside options as well as Fréchet-based structural model of crop choice, we need detailed spatial data on soil suitability for different crops. Standard sources for crop-suitability data are too coarse for our colonies.⁴⁹ We therefore gathered agro-climatic data on the Caribbean at an unusually fine spatial level and then developed agro-climatic suitability indexes for the eight most important crops in our data.

The second source of novel data is archival. Starting in the mid-1830s, Britain began collecting statistics on colonial conditions. Each colony filled out an annual *Blue Book* and sent it to London where it is now stored in the British National Archives. We photographed thousands of the relevant *Blue Book* pages. We also made use of the *Statistical Tables Relating to the Colonial and Other Possessions of the United Kingdom*, annual Censuses, and various other House of Commons parliamentary papers. We manually entered the relevant data into spreadsheets and built a panel data set on exports and export prices by crop, race demographics, wages, incarceration rates, coercive taxes and military expenditures. The panel consists of 14 colonies from 1838 to 1913.

B.1. Wage and Employment Data

Praedial wages: The *Blue Books* report wages for trade, domestic labour and praedial labour (praedial means agricultural). We treat the praedial wage as the wage paid for plantation work. While the *Blue Books* do not explicitly state this, it is implicit from the context, namely, that there was no other agricultural activity that paid wages. Aside from plantation work, the only significant amount of agricultural work was on smallholds and we now provide evidence that smallholds did not hire workers. Eisner (1961) shows that the majority of Jamaican smallholds were under 2 hectares, which even today is considered a very small farm⁵⁰ and hence requires the peasant to search for off-smallhold work, i.e., it was rare for smallholders to hire workers and hence to pay wages.

⁴⁹ For example, each grid cell in the *Geographically Based Economic Data* (GBED) database (e.g., Michalopoulos, 2012) has a resolution of 0.5° latitude by 0.5° longitude, which, at the equator, is over 3,000 square kilometres. Likewise, the crop-suitability data compiled by the FAO GAEZ project (e.g., Costinot *et al.*, 2016) is at the 5 arc-minute level, which, at the equator, is 86 square kilometres. The smallest island in our data, Nevis, is as big as one cell in the FAO GAEZ data. The ten smallest islands in our data together fit into a single cell in the GBED database.

⁵⁰ For example, “‘Profile of the Small-Scale Farming in the Caribbean’, Workshop on Small-Scale Farming in the Caribbean”, Barbara Graham (2012, IMF, page 13).

Table B1. *Partial Correlation between N_{it} and Agricultural Employment.*

Outcome:	(1) Number employed in agriculture/population	(2) Number employed in commerce/population	(3) Number employed in commerce/population	(4) Number employed in commerce/population	(5) Number employed in manufacturing/population	(6) Number employed in manufacturing/population
<i>Panel A: Planter power</i>						
N_{it} : plantation exports as share of total exports	0.3319 [0.0003]		0.0007 [0.9716]		−0.5276 [0.4356]	
N_{it} : sugar exports as share of total exports		0.2392 [0.0002]		−0.0030 [0.8117]		−0.1515 [0.7084]
Observations	125	125	102	102	16	16
R^2	0.381	0.374	0.291	0.293	0.823	0.762
<i>Panel B: Outside options</i>						
$O_i \times$ English exports to all non-Caribbean, HDI_{it}	−0.0955 [0.0058]	−0.1363 [0.0326] 0.1103 [0.2516]	−0.0004 [0.9526]	0.0003 [0.9671] −0.0020 [0.8966]	0.0124 [0.7840]	0.0124 [0.7840]
Observations	125	125	102	102	16	16
R^2	0.311	0.323	0.291	0.291	0.745	0.745

Notes: This table presents results of OLS regressions where the outcome is the total number of people employed in agriculture (columns 1–2), commerce (3–4) and manufacturing (5–6) divided by the total population and the regressor is our N_{it} . With so few observations, we otherwise only include colony fixed effects.

Anecdotal evidence supports the claim the smallholders were subsistence farmers who did not make wage offers. Paget (1964) provides many contemporary quotes from the 1838–40 period. For example: ‘It appears to me that the land which they purchase is . . . too little in extent to be looked to as a permanent source of subsistence and that they must calculate either on obtaining additional means of comfort by going out to labour, or on taking more land on lease’ (Paget, 1964, p. 48). For a later period, the West Indies Royal Commission of 1897 makes the stronger point that, except for plantation work, there were no other wage opportunities in agriculture: ‘If work cannot be found for the labouring population on estates, they must either emigrate or support themselves by cultivating small plots of land on their own account’ (West India Royal Commission, 1897, p. 17).

This discussion establishes that praedial wages were plantation wages and those paid them were either full-time employees or part-timers with subsistence smallholds.

Employment and non-agricultural outside options: Were there other, non-agricultural options for peasants? These appear to have been very limited or non-existent, as suggested by the Royal Commission quote above. More systematic evidence to this effect can be gleaned from the *Blue Books*. The *Blue Books* list three types of employment: agricultural, manufacturing and commercial. The *Books* report that there was virtually no manufacturing employment and commercial employment was typically tiny. The population share of those employed in commerce never exceeded 7%, while that in agriculture was as high as 90% in some colony years.⁵¹ Table B1 shows in columns 1–2 that a complete collapse of the plantation system ($\Delta N_{it} = -1$) was associated with a 33-percentage-point decline in the population share in agricultural employment. This exactly equals the average agricultural employment share. By way of a counterfactual argument, if wage labour on smallholds had been an option for plantation workers,

⁵¹ These data were reported in the population table. However, employment data were much more sparse because they were only occasionally reported or updated. Furthermore, it was reported at the parish level but with year-to-year variation in which parishes were reported so that aggregation to the colony level was frequently not possible.

we might have expected this correlation to be close to zero given as plantation agriculture was replaced by smallhold agriculture. Columns 3–6 of Table B1 also show no significant correlation between N_{it} and employment in commerce or manufacturing. By way of a counterfactual argument, if wage work in the towns had been an option for plantation workers, we might have expected this correlation to be significantly negative.

Thus, the *Blue Books* do not provide any evidence of non-agricultural outside options for plantation workers.

B.2. Export Price Index $r_i(\mathbf{p}_t, \tau^s)$

We used the *Blue Books* to construct a 76-year panel of exports by colony and crop, generating a database containing exports by colony and year for 17 products accounting for 98% of exports.⁵² We then built prices as export unit values (export revenues divided by export quantities) for the 17 products. See Online Appendix C for a description of the price data. We validated our series against the sporadic contemporary sources available. For example, for sugar our export unit values are virtually identical to data in Deerr (1950), the seminal work on the subject, and to sugar prices in Blattman *et al.* (2007).

We now describe how we estimate $r_i(\mathbf{p}_t, \tau^s)$ in equation (1). First, except for sugar, no other crop was exclusively a plantation crop, e.g., coffee was typically grown both on plantations and smallholds. We therefore had to determine for each island whether a crop was fully or partly grown by plantations; see discussion in Online Appendix A. Second, we use the same information on agro-climatic conditions that we utilised for section Appendix B.4 and develop the agro-climatic suitability indexes for the seven most important plantation crops beyond sugar.⁵³

Let A_{gi} be a vector of crop suitability characteristics on island i pertaining to crop g . We estimate the parameters of $r_i(\mathbf{p}_t, \tau^s)$ using an almost standard gravity equation method common to Fréchet-based models. That is, we relate crop-level exports to the $T_{gi}(\tau_g^j)^\theta (p_{gt})^\theta$ terms in equation (1). In the standard gravity approach a plot's productivity is drawn from a single distribution, but here it is drawn from two distributions, depending on whether the plot is a plantation (τ_g^p) or a smallhold (τ_g^s). To keep the estimation within a standard framework we assume that a plot's productivity is drawn from a single distribution with a parameter τ_g which is the geometric average of τ_g^p and τ_g^s . Specifically,

$$\tau_g(Pl_{git}) \equiv (\tau_g^p)^{Pl_{git}} (\tau_g^s)^{1-Pl_{git}} \quad (B1)$$

where Pl_{git} is the share of crop g in colony i in year t produced on plantations.⁵⁴ We also assume that $\ln T_{gi} = \alpha_g A_{gi}$ where α_g is a parameter vector. Gravity estimation very roughly boils down to regressing log exports on A_{gi} , Pl_{git} , a crop dummy, and $\ln p_{gt}$.⁵⁵

⁵² The products are sugar, livestock, arrowroot, cocoa, lime juice, cotton, bananas, oranges, pimento, coffee, charcoal, lumber, coconuts, ginger, other spices (cloves, mace and nutmeg), balata (a natural tar) and asphalt.

⁵³ This entails identifying the agro-climatic factors relevant for each crop. For example, lime has seven factors including average temperature (23–30°C is very suitable) and soil pH (6.1–6.5 is very suitable). For each crop, the relevant factors are ranked in their importance and aggregated into discrete suitability bins that reflect meaningful cutoffs in overall suitability for given crops. The details of our crop suitability coding appear in Online Appendix B and Online Appendix Tables B.1 and B.2.

⁵⁴ This is closely related to the more rigorous solution to this problem in Antràs and de Gortari (2017).

⁵⁵ To see this note that $\ln \tau_g^j = Pl_{git} \ln \tau_g^p + (1 - Pl_{git}) \ln \tau_g^s = \ln(\tau_g^p/\tau_g^s) Pl_{git} + \ln \tau_g^s$. Hence, $\ln[T_{gi}(\tau_g^j)^\theta (p_{gt})^\theta] = \alpha_g A_{gi} + \theta \ln(\tau_g^p/\tau_g^s) Pl_{git} + \theta \ln \tau_g^s D_g + \theta \ln p_{gt}$ where D_g is a crop dummy. This explains the regressors A_{gi} , Pl_{git} , D_g and $\ln p_{gt}$.

Substituting $\hat{T}_g \equiv T_g(\tau_g p_g)^\theta$ into equation (A1) of Appendix A and adding subscripts, the average earnings per plot from crop g are

$$r_{gi}(\mathbf{p}_t, \tau) = [T_{gi}(\tau_g p_{gt})^\theta] [\sum_k T_{ki}(\tau_k p_{kt})^\theta]^{\frac{1}{\theta}-1} \Gamma. \quad (\text{B2})$$

To estimate the parameters of this equation, recall that we assumed $\ln T_{gi} = \alpha_g A_{gi}$ and $\tau_g = \tau_g(P_{l_{git}}) \equiv (\tau_g^P)^{P_{l_{git}}} (\tau_g^S)^{1-P_{l_{git}}}$ where $P_{l_{git}}$ is the share of exports produced on plantations. Then differentiating equation (B2) between crop g and any other crop g' yields

$$\begin{aligned} & \ln r_{gi}(\mathbf{p}_t, \tau) - \ln r_{g'i}(\mathbf{p}_t, \tau) \\ &= \ln [T_{gi}(\tau_g(P_{l_{git}})p_{gt})^\theta] - \ln [T_{g'i}(\tau_{g'}(P_{l_{g'it}})p_{g't})^\theta] \\ &= \alpha_g A_{gi} - \alpha_{g'} A_{g'i} + \theta \ln(\tau_g^P / \tau_g^S) P_{l_{git}} - \theta \ln(\tau_{g'}^P / \tau_{g'}^S) P_{l_{g'it}} + \theta \ln(\tau_g^S / \tau_{g'}^S) + \theta \ln p_{gt} / p_{g't}. \end{aligned}$$

We assume that exports per plot are a noisy measure of output per plot: $\ln x_{git}/n_i = \ln r_{gi} + \varepsilon_{git}$ where n_i is the number of plots. Then we obtain the estimating equation

$$\begin{aligned} \ln(x_{git}/x_{g'it}) = & \beta_g A_{gi} + \beta_{g'} A_{g'i} + \beta'_g P_{l_{git}} + \beta'_{g'} P_{l_{g'it}} + \beta''_g D_g + \beta''' \ln p_{gt}/p_{g't} \\ & + \nu_{git} \quad \forall i, t \text{ and } g \neq g' \end{aligned} \quad (\text{B3})$$

where β s are regression coefficients, g' is fixed (it will be sugar), D_g is a crop dummy and $\nu_{git} \equiv \varepsilon_{git} - \varepsilon_{g'it}$. It is easy to verify that the estimates of equation (B3) identify all the parameters of (B2) except τ_g^S , e.g., $\beta''' = \theta$. Thus, $r_{gi}(\mathbf{p}_t, \tau)$ and $r_i(\mathbf{p}_t, \tau) = \sum_g r_{gi}(\mathbf{p}_t, \tau)$ (see equation (A2)) are known up to the multiplicative constant τ_g^S . Recall that if $P_{l_{git}} = 0$ for all g then $\tau = \tau^S$, i.e., if there is no plantation production, then τ takes on its smallhold value τ^S . Hence, setting $P_{l_{git}} = 0$ for all g , $r_i(\mathbf{p}_t, \tau^S)$ is also known up to the multiplicative constant τ_g^S . In regressions we use $\ln r_i(\mathbf{p}_t, \tau^S)$ so that the additive constant $\ln \tau_g^S$ is subsumed into intercepts.

Equation (B3) is our gravity equation, where g' is sugar. Data on the share of exports produced on plantations ($P_{l_{git}}$) are described in Online Appendix A. Data on agro-climatic conditions (A_{gi}) are described Online Appendix B. The bin-level agro-climatic variables are averaged up to the colony level to create the A_{gi} . Data on export prices (p_{gt}) are described in Online Appendix C. Online Appendix Table C.3 displays the estimates of the key parameters. There are 7,126 = 1,018 $\times$ 7 observations where 1,018 is the number of colony-year pairs and 7 is the number of (non-sugar) crops. We estimate θ to be 2.30 ($t=3.93$). This is in line with Costinot *et al.* (2016) who estimate it to be 2.46.

The remaining estimates reported in Online Appendix Table C.3 are the coefficients on $P_{l_{git}}$. Except for livestock they are all positive as expected. There are several things to note. The negative livestock coefficient is largely driven by a comparison of Virgin Island smallholds with Tobago plantation and it is thus not surprising to find that the Virgin Islands did better. Cotton is always a smallhold crop, which means that its coefficient ($\beta'_g = \theta \ln(\tau_g^P / \tau_g^S)$) and hence τ_g^P are not identified; however, we do not need to know τ_g^P in order to recover $r_i(\mathbf{p}_t, \tau^S)$.⁵⁶ Likewise, the sugar coefficient is not identified because sugar is always a plantation crop. This does not matter because sugar does not enter into the Fréchet smallholder price index. In summary, the gravity

⁵⁶ This is important for another reason: most other smallhold crops (e.g., garden vegetables, roots and tubers) were perishable, and therefore do not appear in any of our export price series. It is worth noting that suitability for smallhold crops other than cotton would not have varied much across islands since these are hardy crops that could grow anywhere in the fertile Caribbean climate. This is witnessed by the fact that they were, in fact, grown everywhere; on plantations as well as in the hinterlands.

equation (B3) identifies the parameters $\{\alpha_g, \tau_g^p, \tau_g^s\}_{g=1}^G$ and θ needed to recover estimates of the $r_i(\mathbf{p}_t, \tau^s)$.

B.3. Coercion Data

In this appendix we address two questions. First, what share of incarcerations or court convictions were associated with legal coercion? Second, what is the relationship between incarcerations and court convictions, given that the latter involved both fines *and* incarcerations?

Incarceration and conviction data were rarely reported in sufficient detail to isolate crimes associated with legal coercion. Fortunately, in the earliest post-abolition years, London was concerned about the abuse of legal coercion by planters and so requested detailed reports on each colony's Acts, incarcerations, convictions and evictions of peasants from plantations. Colonial assemblies did not want to comply and often did not: sometimes there are no tables and sometimes the tables are very imperfect. Where they are available, they are reported in complicated ways. For example, in some colonies there is a table for *each* magistrate separately along with details of *each* case and its outcome (fine, incarceration, eviction). This is a useful source for our fine-grained historical discussion of legal coercion, but not for statistical analysis. Fortunately, in 1838, four of the Leewards (Antigua, St Kitts, Nevis, the Virgin Islands) actually filled in London's standardised and very detailed tables of convictions by detailed crime. These tables appear amid hundreds of pages of detailed exegesis on colonial laws, and as a result the detailed crimes can be fully understood and categorised as coercive or not. The tables were combined into a single table that appears as our Table B2. The first column lists the detailed crimes. The first group ('Coercive') are the crimes that were described in Subsection 1.1 and are clearly related to legal coercion. Note that breach of contract, vagrancy and trespass were all lumped in the Vagrancy Acts of a number of colonies,⁵⁷ which illustrates that one cannot classify crimes as coercive or non-coercive without detailed knowledge of the laws of each colony. Turning to the other coercive crimes in Table B2, licensing was used to limit the opportunities for non-plantation activities such as huckstering (selling goods), jobbing (handyman), boating (transporting goods) and deporting (moving peasants to another island, which was illegal in the Leewards).⁵⁸ Malicious injury to property could stem from the eviction of a peasant from the plantation, the subsequent fight over the peasant's access to his/her crops, and retribution on both sides for perceived injustices.⁵⁹

This discussion of Table B2 explains why it is so difficult to identify legal coercion from the crime statistics. It also makes clear that we can do so for 1838 because we have the supporting Acts and legal discussion of them. Assuming that half of the 'partially coercive' crimes were coercive, 60% of convictions involved legal coercion ($60 = 41 + 37/2$). *Legal coercion was thus an important source of convictions.*⁶⁰

⁵⁷ For example, the Vagrancy Act for St Kitts states that 'all persons shall be deemed vagrants who [are] dwelling in any of the houses upon any plantation without the sanction of the owner or director thereof, or shall be found trespassing on the land of any plantation by attempting to cultivate . . . [or] have left without sufficient cause any work unfinished, which they have contracted to perform' (House of Commons Parliamentary Papers, 1839b, p. 19).

⁵⁸ The restrictive provisions of these Acts were condemned by London-appointed governors, e.g., House of Commons Parliamentary Papers (1839b, p. 41).

⁵⁹ Again, classification is not neat. For example, the Virgin Islands Vagrancy Act covers attempts to burn down plantations (House of Commons Parliamentary Papers 1839b, p. 310) so that malicious injury is sometimes recorded as vagrancy.

⁶⁰ These data are for January to September 1838 and so span 1 August 1838, the end of slavery. This is not an important issue because half of the convictions occurred in Antigua (the most populous colony), which had abolished slavery outright in 1834. (The remaining colonies had abolished slavery, but were in the transition apprenticeship period.)

Table B2. *Convictions (Fines and Incarceration) by Type of Crime.*

	Jan.–Sept. 1838		July–Sept. 1838	
	Number	%	Number	%
Coercive	991	41	425	45
Breach of contract	479	20	227	24
Vagrancies	192	8	91	10
Trespass	137	6	54	6
Hucksters without licence	2	0	0	0
Jobbing without licence	44	2	5	1
Plying unlicensed boat	4	0	0	0
Deportation without licence	4	0	2	0
Malicious injury to property	129	5	46	5
Partially coercive	886	37	348	37
Police Act	257	11	91	10
Riotous, disorderly conduct	77	3	26	3
Assault and battery	552	23	231	25
Non-coercive	516	22	166	18
Larceny	75	3%	22	2
Miscellaneous	441	18%	144	15
Total	2,393	100%	939	100

Notes: Data compiled by authors from House of Commons Parliamentary Papers (1839b), pp. 78–9 (Antigua), 225–6 (St Kitts), 286–7 (Nevis) and 332–4 (Virgin Islands). Hucksters are sellers of small wares. Jobbers are handymen. The Police Act varies from colony to colony. It always covers day-to-day non-coercive offences (e.g., startling a horse) and, in some colonies, offences relating to coercion. Riotous and disorderly conduct ranges from public drunkenness to organised peasant protests. Assault and battery includes assaulting an employer. ‘Miscellaneous’ is the authors’ own category and collects crimes that are not related to coercion of smallholders, e.g., petty theft, mutiny and abusive language. The results are very similar for January–September and July–September. See footnote 60 for a discussion of the differences between these two periods.

We next turn to the question of what percentage of these court convictions involved incarceration. Recall that convictions led to either fines or incarceration. From inspection of convictions tables from 1838, a year in which such tables were collected and published in a major House of Commons report, several patterns emerge. First, the vast majority of higher court convictions involved incarceration, e.g., House of Commons Parliamentary Papers (1839b, p. 72). Second, lower court cases involved either mediation (no conviction, one party agrees to indemnify the other), or conviction with fine, or conviction with imprisonment (for more significant crimes such as long or repeat absenteeism and neglect leading to death of livestock).⁶¹ The only comprehensive account of lower court cases for 1838 is a single table for Barbados (1 August–15 October). In that table, 68% of the convictions involved imprisonment (House of Commons Parliamentary Papers, 1839a, pp. 104–5).

Summarising what we could glean from these reports, the majority of court convictions were for offences that involved legal coercion, and the majority of court convictions resulted in incarceration, albeit usually only for two weeks or one month, often at hard labour.

Erring on the side of caution we also report the results for the last quarter available (July, August and September, 1838). For this period, the percentage of legal coercion rises slightly to 64%.

⁶¹ Note that imprisonment for failure to pay a fine occurred, but this is not the same as saying that a prisoner could buy his or her way out of jail. Sentences were typically of the form ‘sentenced to pay a fine of x dollars; in default, y days’ imprisonment’. In St Kitts in the second half of 1838, convictions for breach of contract/vagrancy resulted in 46 fines and only one of these resulted in imprisonment for failure to pay the fine. See House of Commons Parliamentary Papers (1839b, pp. 200–6).

Table B3. *OLS Regressions of Court Convictions on N_{it} and C_{it} .*

Outcome	(1)	(2)	(3)	(4)	(5)	(6)
	Share of court convictions for offences against property					
N_{it} : sugar exports as share of total exports	0.2471 [0.0372]	0.2204 [0.0898]				
N_{it} : plantation exports as share of total exports			0.3298 [0.0206]	0.3190 [0.0512]		
Incarceration (per cap.)					0.0309 [0.2650]	0.0325 [0.2100]
Time controls	t + t ²	t-fe	t + t ²	t-fe	t + t ²	t-fe
Observations	502	502	502	502	408	408
R ²	0.141	0.227	0.140	0.226	0.136	0.226

Notes: (a) From 1871 to 1913 we observe data on local court sentences by categories of offences. There are four categories: offences against property, offences against the person, 'praedial larceny' (the theft of livestock and crops) and 'other'. Only one of these—offences against property—corresponds closely to the legal coercion discussed in Subsection 1.1. In this table we run our set of three core specifications on three relationships: in columns 1–4, we regress the share of all court sentences that were levied for offences against property, for our two core specifications. In columns 5–6, we regress it on our incarcerations measure. (b) Standard errors are clustered by colony, p -values in square brackets.

For the end of our period (1871–1913) we have an annual panel of court convictions (with no information on what share resulted in incarceration). It would thus be of interest to use these convictions as an alternative measure to legal coercion. To this end, we must identify which convictions are associated with legal coercion. Unfortunately, the conviction data are only available by four broad categories: (i) offences against property; (ii) offences against the person; (iii) praedial larceny (the theft of livestock and crops); (iv) other. The best we can do is equate offences against property with legal coercion. There are pros and cons to this. The *Blue Books* state that offences against property include both offences against rights of property and injuries to the subjects of property, which means that offences against property include trespass.⁶² The *Blue Books* further state that 'by praedial larceny is meant the offence—prevalent in the sugar-growing and Cooley-importing colonies—of robbing provision grounds and homesteads'. By 1871, it probably had only a small component of legal coercion, i.e., of evicted peasants recovering their crops.⁶³ Offences against the person is a category that, as with assault and battery, likely had only a small component of legal coercion, i.e., of peasant retribution against plantation overseers for loss of cottage and crops. 'Other' could subsume any number of offences listed in Table B2 and, since it makes up about 55% of the convictions in our data, it is too much of a catch-all category to be called legal coercion. In summary, we conclude that offences against property are the only category that can be linked to the legal coercion discussed in Subsection 1.1, but this must be done with caution. Note that crimes against property only make up about 15% of the convictions in our data. Table B3 shows that this category's share of all convictions correlates strongly with N_{it} . Specifically, columns 1–4 of the table report a regression of the share of court convictions for offences against property on our two measures of N_{it} . The full specification corresponds to our two baseline specifications in Table 1. There is also a somewhat weaker positive correlation with C_{it} in columns 5–6. We view these patterns as validation of incarceration as our measure of coercion.

⁶² As noted above, breach of contract is not an offence against property.

⁶³ Unfortunately, whether a particular theft was praedial larceny (maximum penalty was a long prison term) or petty theft (maximum penalty was a short prison term) appears to vary across colonies. Petty theft likely falls in the 'other' category.

B.4. Measuring O_i

To measure O_i we carefully calculated the share of each colony's land that is not suitable for sugar cane. This was a major undertaking, but only insofar as the Caribbean islands small size forced us to collect geographic data at an unusually fine level of disaggregation. See Online Appendix B for details.

B.5. Measuring $\bar{W}_t$

Real British exports to non-Caribbean destinations ($\bar{W}_t$) is constructed from data in Mitchell (1988). Real British exports to all destinations are from Table IX.19. These data exclude re-exports.⁶⁴ Nominal British exports to the Caribbean are from Tables IX.5 (1838–47) and IX.16 (1846–1913). Over our period, 83% of British exports were manufactures, so we convert nominal exports to the Caribbean into real exports using the price index for British manufacturing exports. The price index is constructed from data in Tables IX.5 and IX.19. Our instrument $\bar{W}_t$ is real British exports less real British exports to the Caribbean. Data are in millions of 1913 pounds sterling.

B.6. Additional Labour Supply Shocks

There was relatively little cross-island migration in the Caribbean in the nineteenth century. From 1838 to 1913 the most significant movement of people in the Caribbean was the arrival of indentured immigrants from India. We coded up the annualised data as a cumulative stock from Roberts and Byrne (1966). Between 1838 and 1913, cumulative net immigration was 230,000 for Guyana, 124,000 for Trinidad, 37,000 for Jamaica and small amounts for Grenada, St Lucia, Antigua and Dominica. The ratio of cumulative net immigration to 1913 population exceeded 0.15 for only two colonies, Trinidad where it was 0.37 and Guyana where it was 0.77. Restated, immigration was important in only two colonies but in those two it was indeed important. The other labour supply shock was much smaller. British West Indies workers left for Panama during the building of the canal by the French (1881–9) and Americans (1908–13) (Maurer and Yu, 2013, ch. 4). No destination-specific estimates exist of Caribbean emigration but the official numbers for overall emigration reported in the *Colonial Blue Books* were small throughout. We control for the Panama Canal shock with a time dummy for the years in question interacted with inverted distance as a measure of exposure to this shock.

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Online Appendix Replication Package

⁶⁴ Britain imported items such as sugar and then re-exported them.

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